

Optimal online management of vehicle-to-grid charging in corporate parking facilities[☆]

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ABSTRACT

The widespread adoption of electric vehicles (EVs) requires a well-developed charging infrastructure and efficient management strategies. In this context, providing charging opportunities for employees during working hours presents a promising approach. This paper proposes a parking management system for EVs in a corporate setting, designed to optimize the allocation of vehicles to charging stations throughout the workday while managing the power flow to minimize costs and enhance participation in various energy markets. The system addresses real-world challenges, including asynchronous booking requests, stochastic variations in EVs usage, and disparities between the number of charging stations and the demand for charging typically with fewer stations than EVs requiring service. Additional practical constraints are considered, such as variations in charging station characteristics and employees' availability constraints, which may prevent them from plugging in or retrieving their vehicles at certain times. The proposed solution is designed as a service-oriented system capable, at the same time, of dynamically allocating and rescheduling charging sessions in response to user requests. Deterministic data, such as confirmed bookings, and forecast data have been integrated in the proposed solution to improve operational efficiency.

1. Introduction

The broad diffusion of electric vehicles (EVs) has highlighted the critical need for efficient and accessible charging infrastructure [1]. Smart parking systems are emerging as a promising solution in the EV landscape, which allow integration of EV charging and the public infrastructure [2].

Vehicle-to-Grid (V2G) technology enables bidirectional power exchange, converting EVs from passive consumers into active grid participants [3] and supporting power grid stability by mitigating fluctuations and contributing to reserve services, although effectively coordinating EV charging and discharging behaviour for optimal grid participation remains a key challenge [4]. Beyond technical advantages, V2G technology offers economic incentives and supports environmental and social goals through strategic participation in the electricity markets [5–7].

V2G systems contribute to reducing urban congestion, lowering emissions, and promoting sustainable energy practices, benefiting both EV users and urban communities [8]. Optimizing the power management in charging stations and smart parking lots becomes crucial for research and development. Indeed, equipping parking lots with charging points transforms them into dual-function facilities, particularly advantageous in locations with long parking durations, such as office buildings, shopping malls, and airports [9].

However, energy management within these systems is complex due to uncertainties such as arrival and departure times, initial state of charge (SOC), and variable energy consumption [10]. Notably, optimization strategies tackling similar scheduling problems under resource constraints and SOC variability have demonstrated their effectiveness in several other application domains, such as satellite power systems and microgrid energy management [11–13]. Additionally, congestion

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management due to a limited number of charging stations presents another challenge [14,15].

This paper proposes a parking management system (based on a mixed-integer linear programming (MILP) model) for a corporate setting with a limited number of charging stations where employees charge their EVs during work hours; users can book charging slots by specifying arrival and departure times, desired final charge level, and unavailable time slots (e.g., due to meetings or work commitments).

The paper is organized as follows: Section 1 introduces the topic of the paper by highlighting its main contributions and reviewing the relevant literature, while Section 2 presents a possible use case. The modeling of the smart parking system and the cost pricing schemes are detailed in Section 3. Section 4 deals with the proposed optimization problem and the related online management algorithm. Among its aims, it has to be consistent with a known offline management profile. An extension to this is proposed in Section 5, where stochastic EVs behaviours together with deterministic ones are taken into account. The related extended algorithm implementation is also provided. Numerical simulations of our algorithm under different scenarios are presented in Section 6, including a comparison of the performance of our proposed model with a greedy policy. Finally, conclusions, limitations and future works are summarized in Section 7.

Notation: $\mathbb{N}$ and $(\mathbb{R}^+)$ $\mathbb{R}$ are the sets of natural and (non-negative) real numbers, respectively; given a variable $\xi \in \mathbb{R}$, $\xi = \xi(t)$ indicates the value of ξ at the discrete time-step $t \in \mathbb{N}$. With some abuse of notation, $[t_1, t_2]$ with $t_1 \in \mathbb{N}$, $t_2 \in \mathbb{N}$, $t_1 \leq t_2$, indicates the set of consecutive integers from t_1 to t_2 , while $[\xi_1, \xi_2]$ with $\xi_1 \in \mathbb{R}$, $\xi_2 \in \mathbb{R}$, $\xi_1 \leq \xi_2$, indicates the interval from ξ_1 to ξ_2 . The star-superscript notation explicitly refers to an optimal value, while the barred notation indicates a provided given value (despite the fact that it will often be an optimal one in some sense).

1.1. Literature review

This section surveys the most relevant contributions, organised around the key dimensions that distinguish our work: optimization approach, market horizon, treatment of uncertainty, and practical operational constraints.

Research concerning EV charging scheduling and market integration has investigated a spectrum of methods and addressed different practical challenges. User-centric models incorporate drivers economic preferences and quality-of-service (QoS) requirements, providing multi-objective formulations to balance cost and service levels [16]. Metaheuristic algorithms have been proposed to generate reservation-based charging schedules, but their computational complexity often prevents real-time deployment [17]. Energy aggregators operating in wholesale markets must cope with uncertainties in ancillary services and optimize revenue from V2G operations [18]. To enhance forecast-driven decisions, machine learning and deep learning approaches have been extensively explored for day-ahead charging optimization to improve forecast-driven decisions [19,20]. In [19] the authors utilize a deep learning framework for day-ahead optimal charging scheduling in parking lots. In [20] co-optimization algorithms aim to simultaneously reduce energy losses and transformer operating costs in day-ahead planning for plug-in EVs. While machine learning and metaheuristic approaches offer flexibility and adaptability, MILP is distinguished by its capacity to explicitly model complex technical and infrastructure constraints [21]. A MILP-based data-driven framework is proposed in [22] to integrate forecasts for PV and base load, optimizes the number of mobile chargers and their schedules.

Moreover, MILP models are widely used for smart parking systems that integrate charging stations with renewable generation and storage; some works emphasize component sizing and infrastructure design [23,24], whereas others address power management in public parking lots under simplified, pre-assigned EV scenarios [25]. Additional MILP formulations study aggregator policies and photovoltaic-based

fleet charging [26–28], but these contributions focus primarily on energy balancing within the ancillary services market (asm) and neglect practical technical constraints such as ramp rates and minimum delivery durations. More recently [29] studies trading strategies for a fleet of 1000 EVs in the continuous idm, showing that idm adjustments significantly reduce financial risk relative to a only dam strategy; yet, stochastic vehicle arrivals and infrastructure constraints are not considered. Similarly, studies modeling EVs charging and discharging in solar-powered parking lots exclude participation in the day-ahead market (dam) and fail to consider stochastic EV arrivals [30]. Stochastic approaches have been proposed in [31–33]. In [31] the optimal economic management of an EV aggregator using a stochastic p-robust optimization technique, focusing on market price uncertainty but assuming EV availability are examined. The day-ahead scheduling proposed in [32] accounts for uncertainty in wind power generation and arrival SOC of the vehicles and assumes known EVs availability. Closer to our setting, [34] proposes a three-stage real-time energy management system for community parking lots that combines day-ahead scheduling of renewable generation, real-time power correction, and priority-based EV charging. However, vehicle-to-grid capabilities, financial market participation and the allocation of EVs to a limited set of charging stations are not addressed. Stochastic methods for demand estimation are studied in [33] in the context of battery energy storage systems sizing for EVs parking lots. However, the authors do not incorporate real-time operational constraints. Another work modeling the EVs' behaviour uncertainty is [35].

Building on this literature, our work extends the day-ahead optimization framework of [35] by integrating multiple market horizons, i.e., the dam, the asm and the intra-day (idm), handling stochastic and asynchronous EV arrivals, enforcing realistic technical constraints (including ramp rates and minimum delivery durations), and, most importantly, delivering algorithms apt for real-time operations based on online data coming from a realistic corporate parking environments. Table 1 summarizes the comparison with representative prior contributions where the MILP model is used, while in the following subsection we highlight the main contributions of the work.

1.2. Our contribution

When investigating approaches for managing V2G parking systems, realistic scenarios with real world technical constraints and solutions that are implementable in practice are essential. With this in mind, we build on the optimization framework of [35] and deliver a unified, implementable solution that closes gaps left by prior work. Specifically, we

- Extend the optimization model to support multi-horizon market participation (dam, asm and idm). The cost function prioritizes charging in economically optimal time slots while accounting for scheduling on a limited number of charging points and

Table 1

Comparison with prior works, where dam, asm, idm, denote day-ahead, ancillary service and intra-day market respectively (✓ included, ✗ not included).

Ref	dam	asm	idm	QoS	Stochastic Price	Stochastic Time of Arrival	Limited Number of Chargers
[16]	✗	✗	✗	✓	✗	✗	✓
[21,22]	✗	✗	✗	✗	✗	✗	✓
[23]	✗	✓	✗	✗	✗	✗	✗
[28]	✓	✗	✗	✗	✗	✓	✗
[26]	✓	✓	✗	✗	✗	✗	✓
[27,30]	✓	✓	✗	✗	✗	✗	✗
[31]	✓	✓	✗	✗	✓	✗	✗
[35]	✓	✓	✗	✗	✓	✓	✗
[29]	✓	✗	✓	✗	✓	✓	✗
Proposed	✓	✓	✓	✓	✓	✓	✓

promoting market participation. Unlike [35], our approach explicitly integrates intra-day market operations, enabling dynamic energy trading to optimize both charging costs and grid contributions. While [29] also considers idm for EV fleets, it does not address parking allocation with limited chargers, asynchronous user requests, or the joint optimization of charging stations assignment and power management required in a corporate setting.

- Integrate stochastic and asynchronous EV arrivals with a combined scheduling and power-management layer for scarce charging infrastructure. Differently from earlier works, our framework is designed for supporting online, field-deployable operation by accepting asynchronous slot requests, handling unscheduled arrivals, and adapting schedules in real-time while preserving fairness and service guarantees. Moreover, users may declare “do not disturb” intervals during which their vehicle must not be connected or moved from charging stations and the scheduler enforces these constraints when allocating slots and power.
- Enforce practical technical constraints such as ramp rates and minimum delivery durations, and may support heterogeneous chargers.
- Use a number of charging stations lower than the number of EVs requesting charge. This implies that our algorithm must perform, at runtime, both optimal allocation and optimal power profile computation. The fact that the number of stations is less than that of EVs implies that EVs are handled to exploit the charging stations throughout the working day. In contrast, several papers reviewed in the literature consider the same number of EVs and rechargers. Such an assumption might not be met in a company parking lot since it would require higher CAPEX.
- Introduce flexible QoS options that permit controlled deviations from requested SOC, allowing employees to accept or refuse a final SOC that differs from their initial request, thus covering more customers with limited number of chargers.
- Design an online algorithm that exploits a day-ahead offline trajectory. We design a real-time online algorithm that leverages an offline day-ahead trajectory computed on historical/forecast data. The calculation of an offline management trajectory, usually a day in advance, has been addressed in several articles in the literature (see [35]). The online algorithm operates on real data (commonly different from the forecasted data used for the offline solution) attempting to adhere as closely as possible to the offline schedule while accounting for real-time actual data and realistic operating constraints, and remaining fast enough to be offered as an online service. In the first part of the paper (up to Section 4) the day-ahead offline solution is assumed known and only real-time deterministic data are considered for recomputing an online management. In the second part of the paper (Section 5) we extend our approach by integrating forecasted data, i.e., stochastic EV behaviour, with deterministic ones. This extension is also useful for computing offline solutions the day before by powering the online management algorithm.

To summarize the novelty of the paper, from one side we include in the model features that are in general spread/fragmented across different papers in the literature, which deal with few aspects at the time, including multi-market participation, a limited number of recharging spots compared to the EVs, and deterministic and stochastic data simultaneously present in the problem. On the other side, we propose an algorithm able to simultaneously execute scheduling and power balancing, where calculations are compatible with runtime constraints while providing field-deployable solutions that adhere well to offline long time computed (not runtime compatible) optimal solutions. Also, the possibility of flexible QoS is explicitly modelled.

2. Setup description and use case

The parking facility considered in this paper is owned and operated by a private company, that provides employees with an area to recharge

their EVs during the working day. For the sake of generality, it is assumed that flexible working days are allowed, meaning employees can arrive and leave the job at different times during the day. Note that a fixed time can be seen as a particular case of the presented one.

Employees pay for charging their EVs. The corporate sets electricity buying and selling prices daily, purchasing energy to charge EVs and potentially selling surplus energy back to the grid. The company (that is the parking manager algorithm of the company) operates within known electricity price profiles throughout the day, both for purchasing and for selling energy. The parking management system operates across dam, asm and idm, dynamically buying and selling energy to and from EVs during their parking time while also trading energy with the grid. Indeed, the aim of the company is to maximize the profit of the parking facility by leveraging energy market participation. By optimizing energy transactions and scheduling EVs efficiently, the company aims at maximizing its financial return, which could either increase company revenue or be redistributed among employees to incentivize EVs adoption in line with sustainability goals. The setting considers a possibly number of charging parking lots less than the total number of EVs owned by the corporate employees. Since charging stations are expensive, the company seeks to minimize the number of stations while ensuring high utilization. Therefore, rather than operating many underutilized stations for only a few hours daily, the company prioritizes a management strategy that schedules EVs for charging throughout the whole day.

A representative operating scenario is illustrated in Fig. 1, which emphasizes the actions carried out by the online algorithm and by each EV, while dashed lines represent optional actions. A user sends a request and is answered back with either an allocation or a notification stating that

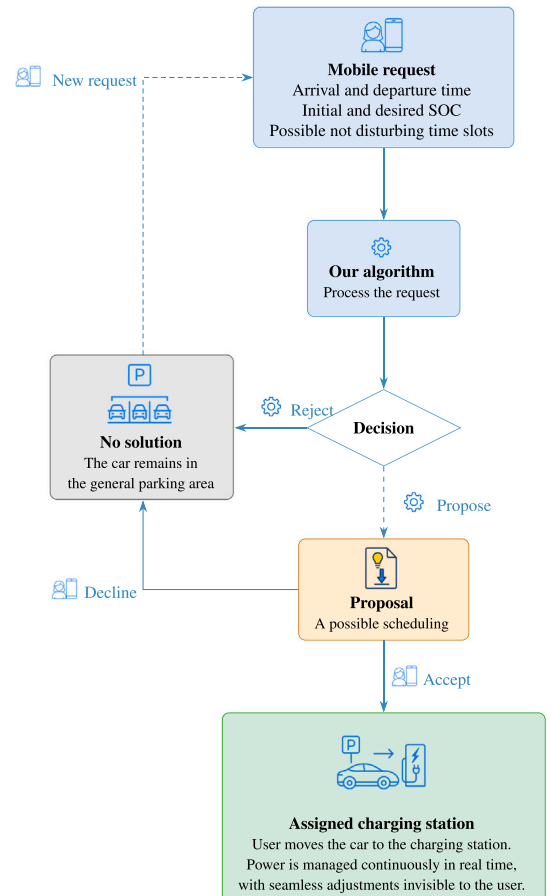


Fig. 1. Smart parking use case.

no feasible allocation was found. The commercial acceptance of the proposed allocation is not addressed within the scope of the management framework. Accordingly, for validation purposes we assume that users for whom an allocation is proposed always accept it and proceed with charging their vehicles. When an employee requires charging for her EV, she informs the parking manager through a dedicated user interface (whose development is outside the scope of this study), specifying:

- Expected arrival time at work.
- Expected departure time.
- Desired SOC range at the end of the charging session (including the option for a full charge).
- Initial SOC upon arrival (either known or estimated by onboard navigation algorithms that provide the arrival time and the expected consumption and/or residual battery charge).
- “Do not disturb” time slots, when she cannot put in charge or remove from charging her vehicle (e.g., due to meetings).

Once the request is submitted, the parking manager may assign a charging station and a corresponding time slot within the employee’s working hours. Charging slots may be reserved in advance (e.g., up to one day prior) or requested on short notice. This flexibility does not affect the performance of the proposed algorithm.

If no feasible charging schedule can accommodate the request, the system notifies the employee of the unavailability.

In addition, the framework includes a reduced QoS option. It is indeed possible to specify a certain amount of energy deviation from the desired final SOC. An employee might decide to accept a proposed allocation with a reduced QoS in change of a discount on the final price. As will be shown later, even a small amount of possible flexibility can be conveniently exploited by the optimization algorithm to further increase the number of customers that can be served and boost the total revenue.

In cases where no solution is possible based on the provided data, the manager informs the employee of the inability to fulfil the booking. Then, the employee may optionally submit a new request with revised parameters.

Once a charging slot is assigned, the employee is required to move her vehicle from the general parking area to the designated charging station at the scheduled time, and to remove it promptly at the end of the session. The assumption that users relocate their vehicles during assigned time slots is consistent with the goal of the paper, which is providing an online algorithm for managing a company parking lot where the users might move the car after having recharged it. Whenever a user cannot move the vehicle in certain time slots (due to uninterruptible working activities) such unavailability time periods can be specified by the user and the allocation solution will respect such constraints. Notice that parking the vehicle in a recharging slot only for the time required for recharging is what actually happens in several European countries, where (public) parking regulations typically permit occupancy of recharging bays only for the duration of active recharge.

A detailed description of the parking model is provided in the following section.

3. Parking model and pricing cost scheme

The considered parking facilities are made up of a set $\mathcal{R} = \{0, 1, \dots, R-1\}$, $R \in \mathbb{N}$, of charging stations, each uniquely identified by an integer. EVs requesting charging slots are represented by the set $\mathcal{V} \subset \mathbb{N}$, with each EV assigned an integer identifier. Given a finite time horizon for the allocation problem, EVs requesting charging outside this interval are assigned new identifiers, i.e., a single physical EV may possess multiple identifiers across different time intervals.

The assignment of i -th EV to charging station $j \in \mathcal{R}$ at the discrete time-step $t \in \mathbb{N}$ is modelled using the binary variable $a_{ij}(t) \in \{0, 1\}$, where $a_{ij}(t) = 1$ indicates an assignment and $a_{ij}(t) = 0$ otherwise. The set of EVs assigned at least once within the interval $[t_1, t_2] \triangleq \{t_1, t_1 +$

$1, \dots, t_2\}$, where $t_1 \leq t_2$, is defined as

$$\mathcal{A}(t_1, t_2) = \{i \in \mathcal{V} : \exists t \in [t_1, t_2], \exists j \in \mathcal{R} \text{ s.t. } a_{ij}(t) = 1\}.$$

Each i -th EV must specify its arrival time $\alpha_i \in \mathbb{N}$ and departure time $\delta_i \in \mathbb{N}$ (with $\alpha_i \leq \delta_i$) when requesting a charging session. The parking management algorithm operates within a bounded allocation interval $[S, F]$, with $S, F \in \mathbb{N}$, $S < F$, where F may represent the end of the workday or a defined horizon length. Within the interval $[S, F]$, the set of EVs requesting charging is denoted by $\mathcal{Q}(S, F) \subseteq \mathcal{V}$ and with $i \in \mathcal{Q}(S, F)$ such that $S \leq \alpha_i \leq F$ and $S \leq \delta_i \leq F$.

This set is disjoint from the set of EVs already allocated, $\mathcal{A}(S, F)$, (that is $\mathcal{Q}(S, F) \cap \mathcal{A}(S, F) = \emptyset$), since an already assigned EV cannot request another assignment within the same interval. Both $\mathcal{Q}(S, F)$ and $\mathcal{A}(S, F)$ are assumed to be finite.

The following subsections detail the constraints that ensure the feasibility of vehicle allocation and the optimization of energy consumption and resource allocation.

Vehicles charging power variables

Let us denote with $P_{ij}^+(t) \in \mathbb{R}^+$ ($P_{ij}^-(t) \in \mathbb{R}^+$) the power flowing from (to) the charging station j to (from) the i -th vehicle at time t . These power flows are constrained by the following limits:

$$0 \leq P_{ij}^+(t) \leq z_{i,\bullet}^{+, \max}, \quad (1a)$$

$$0 \leq P_{ij}^-(t) \leq z_{i,\bullet}^{-, \max}, \quad (1b)$$

$\forall j \in \mathcal{R}, \forall i \in \mathcal{Q}(S, F) \cup \mathcal{A}(S, F)$ and $t \in [S, F]$, where $z_{i,\bullet}^{+, \max}$ ($z_{i,\bullet}^{-, \max}$) indicates the maximum actual power flowing in (out of) the i -th EV. Power exchange between the i -th vehicle and station j at time t can only occur if the vehicle is assigned to the charging station, i.e., $a_{ij}(t) = 1$. To enforce this, we define the actual power entering and leaving the i -th vehicle as follows:

$$z_{ij}^+(t) = P_{ij}^+(t)a_{ij}(t) \text{ and } z_{ij}^-(t) = P_{ij}^-(t)a_{ij}(t).$$

These expressions imply that no power is transferred when $a_{ij}(t) = 0$, yielding the following mixed-integer linear constraints:

$$\begin{cases} z_{ij}^+(t) \geq 0, \\ z_{ij}^+(t) \leq a_{ij}(t)z_{i,\bullet}^{+, \max}, \\ z_{ij}^+(t) \geq P_{ij}^+(t) - (1 - a_{ij}(t))z_{i,\bullet}^{+, \max}, \\ z_{ij}^+(t) \leq P_{ij}^+(t), \end{cases} \quad (2)$$

and

$$\begin{cases} z_{ij}^-(t) \geq 0, \\ z_{ij}^-(t) \leq a_{ij}(t)z_{i,\bullet}^{-, \max}, \\ z_{ij}^-(t) \geq P_{ij}^-(t) - (1 - a_{ij}(t))z_{i,\bullet}^{-, \max}, \\ z_{ij}^-(t) \leq P_{ij}^-(t), \end{cases} \quad (3)$$

$\forall j \in \mathcal{R}, \forall i \in \mathcal{Q}(S, F) \cup \mathcal{A}(S, F)$, and $t \in [S, F]$.

Each vehicle can be connected to at most one station that will be assigned by our proposed management algorithm as we will detail below. Therefore, with the assignment being unknown at this stage, for convenience, we define the following variables:

$$z_{i,\bullet}^+(t) = \sum_{j \in \mathcal{R}} z_{ij}^+(t), \quad (4)$$

and

$$z_{i,\bullet}^-(t) = \sum_{j \in \mathcal{R}} z_{ij}^-(t), \quad (5)$$

$\forall i \in \mathcal{Q}(S, F) \cup \mathcal{A}(S, F)$ and $t \in [S, F]$.

Vehicle power rate constraints

To protect battery life, the power entering and leaving each vehicle is subject to ramp rate constraints, limiting sudden changes in power flow. These constraints ensure smooth power variations over time.

$$z_{i,\bullet}^+(t+1) - z_{i,\bullet}^+(t) \leq G_i^+, \quad (6a)$$

$$z_{i,\bullet}^+(t+1) - z_{i,\bullet}^+(t) \geq -G_i^+, \quad (6b)$$

$$z_{i,\bullet}^-(t+1) - z_{i,\bullet}^-(t) \leq G_i^-, \quad (6c)$$

$$z_{i,\bullet}^-(t+1) - z_{i,\bullet}^-(t) \geq -G_i^-, \quad (6d)$$

$\forall i \in \mathcal{Q}(S, F) \cup \mathcal{A}(S, F)$ and $t \in [S, F-1]$, where G_i^+ and G_i^- represent the ramp rates for vehicle i when receiving and supplying power, respectively. These constraints prevent excessive variations in charging and discharging power over consecutive time steps.

Vehicle assignment constraints

Each vehicle requesting a charging slot can be assigned to at most one charging station at any given time

$$\sum_{j \in \mathcal{R}} a_{ij}(t) \leq 1, \quad (7)$$

$\forall i \in \mathcal{Q}(S, F)$ and $t \in [S, F]$. This constraint ensures that an EV is never assigned to multiple charging stations simultaneously. Note that this condition applies only to vehicles requesting charging slots, i.e., EVs included in $\mathcal{Q}(S, F)$, and not to those already assigned, i.e., EVs included in $\mathcal{A}(S, F)$. The latter, in fact, must preserve their pre-scheduled allocations to avoid disrupting employee schedules. This is enforced by the following constraint:

$$a_{ij}(t) = \bar{a}_{ij}(t), \quad (8)$$

$\forall i \in \mathcal{A}(S, F)$, $\forall j \in \mathcal{R}$ and $t \in [S, F]$, where $\bar{a}_{ij}(t)$ represents the previously determined assignment status. This assures that assigned vehicles maintain their scheduled slots, preventing any changes that could disrupt previously planned allocations.

To prevent scenarios where a vehicle temporarily leaves a charging slot and later re-parks, the assigned charging period must consist of consecutive time slots within the interval $[S, F]$. This is enforced through the following constraints:

$$a_{ij}(t) - a_{ij}(t-1) + 1 \geq a_{ij}(h), \quad h = t+1, \dots, F, \quad (9a)$$

$$a_{ij}(t) - a_{ij}(t-1) + 1 \geq a_{ik}(p), \quad p = t, \dots, F, \quad k \in \mathcal{R} - \{j\}, \quad (9b)$$

$\forall j \in \mathcal{R}$, $\forall i \in \mathcal{Q}(S, F)$ and $t \in [S+1, F]$. Each vehicle can only be assigned a charging slot within its arrival and departure window, bringing to

$$a_{ij}(t) = 0, \quad (10)$$

$\forall i \in \mathcal{Q}(S, F)$ and $t \in [S, \alpha_i - 1] \cup [\delta_i + 1, F]$. This guarantees that no vehicle is scheduled for charging before its arrival time α_i or after its departure time δ_i .

Employees may specify *unavailability periods* during which they cannot move their vehicle (e.g., due to meetings). For each i -th vehicle let $\mathcal{D}_i \subset [\alpha_i, \delta_i]$, denote the set of unavailable time slots. The following constraint ensures that charging assignments remain unchanged during these intervals

$$a_{ij}(t) = a_{ij}(t-1), \quad (11)$$

$\forall i \in \mathcal{Q}(S, F)$ and $t \in \mathcal{D}_i$, where the initial charging status is set as $a_{ij}(\alpha_i - 1) = 0$. This prevents the system from scheduling charging operations at times when the vehicle cannot be moved.

Constraints (6)–(11) collectively define a feasible and structured charging schedule, balancing efficient resource allocation, user-defined constraints, and grid stability while ensuring compliance with operational requirements.

SOC constraints

The battery model includes the charging and discharging dynamics as in [35]. The evolution of the SOC, $e_i \in [0, 1] \subset \mathbb{R}^+$ for the i -th vehicle is expressed as

$$e_i(t+1) = \lambda_i e_i(t) + \tau \left[\eta_i^+ z_{i,\bullet}^+(t) - \frac{1}{\eta_i^-} z_{i,\bullet}^-(t) \right], \quad (12)$$

$\forall i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)$ and $t \in [\alpha_i, \delta_i]$, where τ is the sampling period, $e_i(\alpha_i) = e_{i,\alpha_i}$ represents the initial battery SOC upon arrival, $\lambda_i \in (0, 1]$ accounts for the self-discharging losses and $\eta_i^+, \eta_i^- \in (0, 1]$ are the charging/discharging efficiencies.

To ensure feasibility, the SOC always remains within valid operational limits

$$e_i(t) \geq e_i^{\min}, \quad e_i(t) \leq e_i^{\max}, \quad (13)$$

with $e_i^{\min} < e_i^{\max}$, $\forall i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)$ and $t \in [\alpha_i, \delta_i]$. At departure, the vehicle SOC should ideally fall within the desired SOC interval specified by the owner, denoted as $[e_{i,\delta_i}^-, e_{i,\delta_i}^+]$. Nevertheless, to increase scheduling flexibility, the parking manager may propose a reduced QoS by allowing some relaxation of these SOC bounds. The following constraints codify this feature:

$$e_i(\delta_i) \leq e_{i,\delta_i}^+ + y_{\delta,i}^+, \quad (14a)$$

$$e_i(\delta_i) \geq e_{i,\delta_i}^- - y_{\delta,i}^-, \quad (14b)$$

$$y_{\delta,i}^+ \geq 0, \quad (14c)$$

$$y_{\delta,i}^- \geq 0, \quad (14d)$$

$$y_{\delta,i}^+ \leq y_{\delta,i}^{+, \max}, \quad (14e)$$

$$y_{\delta,i}^- \leq y_{\delta,i}^{-, \max}, \quad (14f)$$

$\forall i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)$, where the non-negative variables $y_{\delta,i}^+, y_{\delta,i}^-$ represent potential deviations from the requested SOC range and the constants $y_{\delta,i}^{+, \max}, y_{\delta,i}^{-, \max}$ indicate the maximum allowed QoS flexibility. Setting $y_{\delta,i}^{+, \max} = y_{\delta,i}^{-, \max} = 0$ ensures that the final SOC must exactly match the requested interval. In contrast, setting both values to a sufficiently large constant (e.g., greater than $e_i^{\max}$) allows total relaxation, expanding the solution space. This enables a trade-off between feasibility and user satisfaction, where the system can suggest minor deviations to improve overall scheduling efficiency while keeping the employee informed of any SOC reductions.

Charging station constraints

In the general setting we consider, a charging station j can serve up to N_j vehicles simultaneously. This is encoded as

$$\sum_{i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)} a_{ij}(t) \leq N_j, \quad (15)$$

$\forall j \in \mathcal{R}$ and $t \in [S, F]$. Notice that while some advanced charging station support multiple concurrent connections, most existing infrastructures can only charge a vehicle at the time. This is obviously represented by operating with $N_j = 1$. Besides constraint (15), each charging station has a maximum power that can be delivered (taken) to (from) the vehicles, denoted as $z_{\bullet,j}^{+, \max} \in \mathbb{R}^+$ ($z_{\bullet,j}^{-, \max} \in \mathbb{R}^+$), respectively. The total power supplied or extracted at any given time must satisfy the following constraints:

$$\sum_{i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)} z_{ij}^+(t) \leq z_{\bullet,j}^{+, \max}, \quad \sum_{i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)} z_{ij}^-(t) \leq z_{\bullet,j}^{-, \max}, \quad (16)$$

$\forall j \in \mathcal{R}$ and $t \in [S, F]$.

Certain charging stations may be unavailable at specific times due to maintenance or other operational constraints. Let $\mathcal{U}_j(S, F) \subseteq [S, F]$

denote the set of time slots during which station j is unavailable. To ensure that no vehicle is assigned to an unavailable charging station, the following constraint is imposed:

$$a_{ij}(t) = 0, \quad (17)$$

$$\forall j \in \mathcal{R}, \forall i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F) \text{ and } t \in \mathcal{U}_j(S, F).$$

Power grid connection

The parking manager participates in idm, dam and asm. The manager can either buy or sell energy in these markets [35]. Therefore, we consider the powers bought ($P_{\text{idm}}^+(t)$, $P_{\text{idm}}^-(t)$, $P_{\text{dam}}^+(t)$) and sold ($P_{\text{dam}}^-(t)$, $P_{\text{asm}}^+(t)$, $P_{\text{asm}}^-(t)$) in these markets at time t . Within the time interval $[S, F]$, the following constraints are given for the idm and dam:

$$P_{\text{idm}}^+(t) \geq 0, \quad (18a)$$

$$P_{\text{idm}}^-(t) \geq 0, \quad (18b)$$

$$P_{\text{dam}}^+(t) \geq 0, \quad (18c)$$

$$P_{\text{dam}}^-(t) \geq 0. \quad (18d)$$

The asm differs from the other markets as it involves real-time power adjustments in response to grid needs. To do so, the parking manager offers power flexibility through upward (selling power) and downward (buying power) adjustments. Therefore, within the time interval $[S, F]$, we have the following constraint:

$$s^+(t) \geq 0, \quad (19a)$$

$$s^+(t) \leq s^{+, \max}, \quad (19b)$$

$$s^-(t) \geq 0, \quad (19c)$$

$$s^-(t) \leq s^{-, \max}, \quad (19d)$$

where $s^-(t)$ and $s^+(t)$ are the upward and downward adjustments, and $s^{+, \max} \in \mathbb{R}^+$ and $s^{-, \max} \in \mathbb{R}^+$ are the relative maximum values. The actual participation in the ancillary market by the parking manager at time t is denoted by the boolean variable $m(t) \in \{0, 1\}$, where true value stands for participation, and false otherwise. Therefore, a power flexibility variation ($z_{\text{asm}}^-(t)$, $z_{\text{asm}}^+(t)$) can only be offered when the (time) corresponding boolean participation variable is true, that is $z_{\text{asm}}^-(t) = s^-(t)m(t)$ and $z_{\text{asm}}^+(t) = s^+(t)m(t)$. These are rewritten as the following mixed integer linear constraints:

$$\begin{cases} z_{\text{asm}}^+(t) \geq 0, \\ z_{\text{asm}}^+(t) \leq m(t)s^{+, \max}, \\ z_{\text{asm}}^+(t) \geq s^+(t) - (1 - m(t))s^{+, \max}, \\ z_{\text{asm}}^+(t) \leq s^+(t), \end{cases} \quad (20)$$

and

$$\begin{cases} z_{\text{asm}}^-(t) \geq 0, \\ z_{\text{asm}}^-(t) \leq m(t)s^{-, \max}, \\ z_{\text{asm}}^-(t) \geq s^-(t) - (1 - m(t))s^{-, \max}, \\ z_{\text{asm}}^-(t) \leq s^-(t), \end{cases} \quad (21)$$

Following [35], the downward (power buying) and upward (power selling) ancillary service participation is given by

$$P_{\text{asm}}^+(t) = z_{\text{asm}}^+(t)\omega^+(t), \quad P_{\text{asm}}^-(t) = z_{\text{asm}}^-(t)\omega^-(t), \quad (22)$$

where $\omega^-(t)$, $\omega^+(t) \in [0, 1] \subset \mathbb{R}^+$ are the downward and upward service signals provided by the transmission system operator (TSO). Since the TSO determines $\omega^+(t)$ and $\omega^-(t)$ in real-time, these values are unknown when the parking manager submits its ancillary service profile.

In some countries, the asm participation further requires a minimum size of consecutive time instants, namely T_{asm} at which the flexibility must be guaranteed. In our model, this constraint is given by

$$m(t) - m(t-1) \leq m(h), \quad (23)$$

with $h = t+1, \dots, \min\{t+T_{\text{asm}}-1, F\}$, $\forall t \in [S+1, F]$.

In view of the fact that the parking facility is connected to a single grid dispatch point, the total power balance, within the time interval $[S, F]$ must be satisfied at each time instant

$$\sum_{i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)} \left[z_{i, \bullet}^+(t) - z_{i, \bullet}^-(t) \right] - P_{\text{idm}}^+(t) + P_{\text{idm}}^-(t) - P_{\text{dam}}^+(t) + P_{\text{dam}}^-(t) - P_{\text{asm}}^+(t) + P_{\text{asm}}^-(t) = 0. \quad (24a)$$

$$P_{\text{idm}}^+(t) + P_{\text{dam}}^+(t) + P_{\text{asm}}^+(t) \leq P^{+, \max}, \quad (24b)$$

$$P_{\text{idm}}^-(t) + P_{\text{dam}}^-(t) + P_{\text{asm}}^-(t) \leq P^{-, \max}. \quad (24c)$$

Since participation in dam and in asm must be pre-decided prior to online scheduling, the following values are fixed during real-time decision-making within the time interval $[S, F]$

$$P_{\text{dam}}^+(t) = \bar{P}_{\text{dam}}^+(t), \quad (25a)$$

$$P_{\text{dam}}^-(t) = \bar{P}_{\text{dam}}^-(t), \quad (25b)$$

$$s^+(t) = \bar{s}^+(t), \quad (25c)$$

$$s^-(t) = \bar{s}^-(t), \quad (25d)$$

$$m(t) = \bar{m}(t). \quad (25e)$$

Remark 1. The constraints provided in this section do not explicitly prevent power from simultaneously flowing into and out of the grid. Likewise (12) does not inherently exclude scenarios where a vehicle is both charging and discharging at the same time. However, such cases are physically meaningless and must be avoided. A straightforward approach to prevent these unrealistic situations would be to introduce mutual exclusivity constraints for power inflow and outflow. However, the structure of the optimization problem itself excludes these solutions, eliminating the need for additional constraints. A detailed explanation is provided in [Appendix A](#).

Having provided the parking modeling, we now describe the costs characterizing the markets where the parking facility plays. These costs will be instrumental in defining one of the two optimization functions characterizing the runtime algorithm. Specifically, the pricing model follows [35], with the extension to the idm. At each time instant t , the *unit energy price* paid by an EV to the parking manager for charging is denoted as $c_v^+(t) \in \mathbb{R}^+$. Conversely, when the parking manager buys energy from an EV (i.e., vehicle-to-grid discharging), the unit price paid to the EV is $c_v^-(t) \in \mathbb{R}^+$. The unit prices for energy purchased from the grid through different markets are indicated with the real positive variables $c_{\text{idm}}^+(t)$, $c_{\text{dam}}^+(t)$ and $c_{\text{asm}}^+(t)$ respectively, while, the prices at which the parking manager sells energy back to the grid are $c_{\text{idm}}^-(t)$, $c_{\text{dam}}^-(t)$ and $c_{\text{asm}}^-(t)$. These energy prices contribute to the objective function of the optimization problem formulated in the next sections. For a realistic and meaningful energy market these prices must satisfy the following relationships:

$$\begin{aligned} c_{\text{idm}}^-(t) &< c_{\text{dam}}^-(t) < c_{\text{asm}}^-(t) < c_v^-(t) < c_v^+(t) < c_{\text{asm}}^+(t) < c_{\text{dam}}^+(t) < c_{\text{idm}}^+(t), \\ c_{\text{dam}}^-(t) &< c_{\text{asm}}^-(t) < c_v^-(t), \\ c_{\text{dam}}^+(t) &< c_{\text{asm}}^+(t) < c_v^+(t). \end{aligned} \quad (26)$$

Notice that, these pricing relationships characterize electricity markets in their normal day life conditions (see for example the Italian case we used to validate our approach [36]) and are structural conditions

chosen to ensure economic consistency of multi market participation and to prevent loss making round trip arbitrage when discharging occurs. Indeed, these values reflect the fact that the idm, dam and asm constitute a hierarchy of increasing volatility. While the idm tends to be more stable than the other two, the latter show greater price variability because of their operational role in balancing and real-time dispatching.

The hierarchy of prices mirrors this level of uncertainty and ensures that energy arbitrage activities remain economically sustainable.

Moreover, to ensure that borrowing energy from a vehicle and returning it later does not result in a financial loss for the EV owner, the pricing scheme must satisfy the constraint

$$c_v^-(t) > \max_{i \in \mathcal{V}, t' \in [S, F]} \frac{1}{\eta_i^+ \eta_i^-} c_v^+(t'). \quad (27)$$

Therefore, these inequalities are inherent to any well-functioning energy market and serve to preclude anomalous practices, such as avoiding participation in the riskier dam and asm, engaging in cross-market arbitrage by buying energy in one market and reselling it in another at a higher price, or deliberately discharging an EV battery to sell the energy to the market or back to the same vehicle owner.

Vehicles recharging cost function

The total cost of energy transactions at each time t includes multiple components, accounting for EVs charging/discharging, market participation, and QoS reductions. The cost (or revenue, if negative) associated with charging and discharging EVs at time t is

$$C_v(t) = c_v^-(t)\tau \sum_{i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)} z_{i,\bullet}^-(t) - c_v^+(t)\tau \sum_{i \in \mathcal{A}(S, F) \cup \mathcal{Q}(S, F)} z_{i,\bullet}^+(t). \quad (28)$$

where τ is the sampling period. This expression captures the revenue from EV discharging ($z_{i,\bullet}^-(t)$), i.e., energy sold by vehicle to the parking manager at price $c_v^-(t)$ and the cost of EV charging ($z_{i,\bullet}^+(t)$), i.e., energy supplied by the parking manager to vehicle at price $c_v^+(t)$.

Market participation cost functions

The cost (or revenue) for participating in the idm per unit of time is

$$C_{\text{idm}}(t) = c_{\text{idm}}^+(t)\tau P_{\text{idm}}^+(t) - c_{\text{idm}}^-(t)\tau P_{\text{idm}}^-(t). \quad (29)$$

The cost (or revenue) for participating to the dam per unit of time is

$$C_{\text{dam}}(t) = c_{\text{dam}}^+(t)\tau P_{\text{dam}}^+(t) - c_{\text{dam}}^-(t)\tau P_{\text{dam}}^-(t). \quad (30)$$

Analogously to the case of the idm and dam, the cost (or revenue) from participation in asm is

$$C_{\text{asm}}(t) = c_{\text{asm}}^+(t)\tau P_{\text{asm}}^+(t) - c_{\text{asm}}^-(t)\tau P_{\text{asm}}^-(t). \quad (31)$$

QoS reduction cost function

To discourage excessive deviations from the originally requested SOC levels, a penalty cost is introduced when the parking manager applies QoS reductions. The associated cost is given by

$$C_y = \sum_{i \in \mathcal{Q}(S, F)} w_{y,i}^+ y_{\delta,i}^+ + \sum_{i \in \mathcal{Q}(S, F)} w_{y,i}^- y_{\delta,i}^-, \quad (32)$$

where $y_{\delta,i}^+$ represents the increase in the final SOC beyond the originally requested maximum, $y_{\delta,i}^-$ represents the decrease in the final SOC below the originally requested minimum, and $w_{y,i}^+$ and $w_{y,i}^-$ are the corresponding penalty coefficients applied to these deviations. By incorporating this penalty, the optimization framework encourages adherence to the original SOC requests while allowing limited flexibility when necessary.

Overall power cost function

The constraints previously described in this section are essential for computing the total power cost, which is used in one of the two optimization problems formulated later in this work. For such an optimization problem, the overall allocation cost for a given scheduling interval $[S, F]$ is given by

$$J_c = \sum_{t=S}^F [C_v(t) + C_{\text{idm}}(t) + C_{\text{dam}}(t) + C_{\text{asm}}(t)] + C_y. \quad (33)$$

This cost function accounts for the energy transactions related to EVs charging/discharging, participation in idm, dam, and asm, and any penalties associated with QoS reductions.

Deviation from the offline scenario cost function

In this part of the manuscript it is assumed that a day-ahead solution has already been pre-computed over historical/forecasted data. An approach for solving this offline problem is described in Section 5. As stated, we consider here, instead, already having offline results and we investigate how the real-time (run-time) problem should incorporate them.

The runtime scheduling algorithm seeks to remain as close as possible to the offline solution while accounting for discrepancies between precomputed forecast-based data and real-time data reflecting actual conditions. To enforce this adherence, we introduce the following deviation cost function

$$J_a = \sum_{t=S}^F \sum_{i \in \mathcal{Q}(S, F)} \sum_{j \in \mathcal{R}} w_{ij}(t) a_{ij}(t). \quad (34)$$

The weight coefficients $w_{ij}(t)$ are designed to penalize deviations from the offline solution, meaning that allocation $a_{ij}(t)$ that differs from pre-computed assignments incurs higher costs. While the specific method for selecting $w_{ij}(t)$ does not affect the optimization framework, a simple yet effective choice is considering $w_{ij}(t) = -1, \forall j \in \mathcal{R}$ if the i -th vehicle is allocated at time t (irrespective of which specific charging station) and $w_{ij}(t) = 0, \forall j \in \mathcal{R}$ otherwise. This selection promotes the tendency of the real-time algorithm to maintain the same time slots identified in the offline schedule while allowing the possibility of selecting alternative charging stations, if needed. This choice is used for the numerical validation discussed later in this work.

4. Optimal formulation and runtime management algorithm

We solve the runtime problem through a two-stage sequential optimization: stage 1 aligns real-time allocation with the offline schedule, while stage 2 minimizes total power cost given the stage 1 allocation.

Stage 1: optimal deviation from the offline solution

The first optimization problem assigns new charging slots for the set of incoming vehicles $\mathcal{Q}(S, F)$, while ensuring the previously allocated vehicles $\mathcal{A}(S, F)$ retain their pre-scheduled charging sessions. The objective is to minimize the deviation cost (34) by solving the following problem:

$$\begin{aligned} & \min J_a \\ & \text{s.t.} \\ & \text{Vehicles charging power variables (1)–(5),} \\ & \text{Vehicles power ramp rate constraints (6),} \\ & \text{Vehicles assignment constraints (7)–(8),} \\ & \text{SOC constraints (12)–(14),} \\ & \text{Charging station constraints (15)–(17),} \\ & \text{Power grid connection (18)–(25).} \end{aligned} \quad (35)$$

Notice that, after having solved problem (35), each i -th vehicle is either successfully assigned a charging slot or remains unassigned if no feasible solution exists. When assigned, the vehicle is allocated to a unique interval $[\tilde{\alpha}_i, \tilde{\delta}_i] \subseteq [\alpha_i, \delta_i]$ consisting of consecutive time slots, ensuring continuity in the charging process.

Stage 2: optimal power solution

Suppose that problem (35) has been solved and that J_a^* denotes the corresponding optimal cost obtained by considering the sets $Q(S, F)$ and $\mathcal{A}(S, F)$. Using the same sets, the following optimization problem is formulated to exploit the available degrees of freedom within the allocation feasibility space in order to minimize the overall power cost:

$$\begin{aligned} \min & J_c \\ \text{s.t.} & \\ & (1)-(25), \\ & J_a \leq J_a^*. \end{aligned} \quad (36)$$

This formulation ensures that, while minimizing power costs, the optimization does not introduce excessive deviations from the offline schedule computed in stage 1.

Key observations include *online data compensation* where, unlike the offline problem that relies on forecasts, both (35) and (36) operate on real-time data, naturally correcting deviations between predicted and actual conditions to improve allocation accuracy. Additionally, *flexibility in cost function design* allows problem (36) to be further extended with additional objectives, such as charging station load balancing to ensure uniform ageing of equipment. This two-stage approach effectively balances adherence to the offline schedule with cost efficiency, providing a practical, online solution for managing EV charging in a dynamic corporate environment.

The optimization problems (35) and (36) are encapsulated in a run-time algorithm detailed in what follows. All the requests are collected in a FIFO queue, say C . When a customer requests for a charging time slot, she provides her arrival and departure times $[\alpha_i, \delta_i]$, possible unavailability periods D_i , the initial and the final range of SOC, i.e., e_{i,α_i} and $[e_{i,\delta_i}^-, e_{i,\delta_i}^+]$, and the possible allowed flexibility, $y_{\delta,i}^{-,\max}$ and $y_{\delta,i}^{+,\max}$.

We assume requests occur at distinct time instants, as the cloud-based queue C processes arrivals so rapidly that simultaneous bookings are practically impossible. In rare cases of concurrency, a random ordering is applied.

The algorithm is executed any time a triggering event is detected. This corresponds to the fact that queue C is not empty, i.e., at least one EV is requesting a recharge. A high-level description of the process involved is reported in Algorithm 1. We report as output only the arrival and the departure time of each i -th EV. Indeed the booking is notified to the customer only with park and pick up times. Power profiles are instead used “internally” to provide the references, that each charging station has to follow. These reference values may be updated dynamically each time Algorithm 1 is executed, but such adjustments occur transparently and do not affect the EV owner’s experience.

5. Scenario approach and combined offline/online management algorithm

The optimization problems (35) and (36) are instrumental for the real-time management in Algorithm 1.

It is therefore assumed that historical data on working habits are available and a day-ahead offline solution is available too. The online management algorithm, based on actual data, tries to adhere as much as possible to the offline solution while satisfying all the constraints and maximizing the profitability.

Algorithm 1: Online parking manager.

Input: $C, S \in \mathbb{R}, F \in \mathbb{R}$
Output: $\mathcal{A}(S, F), Q(S, F), \tilde{\alpha}_i, \tilde{\delta}_i$
 $Q(S, F) \leftarrow \emptyset$;
for $i \in C$ **do**
 $Q(S, F) \leftarrow Q(S, F) \cup \{i\}$;
 $C \leftarrow C \setminus \{i\}$;
 //solve the optimization problem;
 Solve (35);
 if (35) *is feasible* **then**
 Solve (36) which provides $[\tilde{\alpha}_i, \tilde{\delta}_i]$;
 if EV i *accepts* $[\tilde{\alpha}_i, \tilde{\delta}_i]$ **then**
 $\mathcal{A}(S, F) \leftarrow \mathcal{A}(S, F) \cup \{i\}$;
 end
 else
 Notify to the EV i the allocation fail;
 end
 $Q(S, F) \leftarrow Q(S, F) \setminus \{i\}$;
end

In this section we change the viewpoint and focus not only on deterministic aspects but also on optimizing the EVs allocation according to statistical data. We therefore provide a further contribution where the online setting is modified to include both deterministic and stochastic data and optimize over an average scenario realization.

Specifically, together with the deterministic set $Q(S, F)$ of vehicles requesting allocation and those already allocated (set $\mathcal{A}(S, F)$), we introduce an additional finite set of vehicles, denoted as $Q'(S, F)$. This set represents vehicles that are expected to request charging within the time interval $[S, F]$ but have not yet submitted a request at the time the optimization problem is executed. Set $Q'(S, F)$ models historically forecasted employees who are expected to ask to recharge their EVs. The three sets are mutually exclusive, ensuring that $Q(S, F) \cap Q'(S, F) = \emptyset$ and $\mathcal{A}(S, F) \cap Q'(S, F) = \emptyset$.

Since vehicles in $Q'(S, F)$ have not formally requested charging, their characteristics, such as arrival and departure times α_i and δ_i , unavailability periods D_i , initial SOC $e_i(\alpha_i)$, and desired SOC range $[e_{i,\delta_i}^-, e_{i,\delta_i}^+]$, are modelled as independent stochastic variables, each characterized by a known probability density function. Because the initial SOC is stochastic, the battery dynamics in (12) also become a stochastic process, meaning that $e_i(t)$ itself is a stochastic variable.

To differentiate between different realizations of these stochastic variables, we use scenario indexing. Specifically, for each vehicle $i \in Q'(S, F)$, we denote by i^k the k -th realization of the i -th vehicle, meaning that α_{ik} represents the arrival time of the i -th EV in the k -th scenario. The same notation applies to all other stochastic variables.

Since the downward and upward service signals $\omega^-(t)$ and $\omega^+(t)$ are unknown at the time the optimization problem described later is run, they are treated as stochastic variables here as well.

To manage the parking in such stochastic setting, and to compute the decision variables $s^-(t), s^+(t)$ that we considered given in problems (35), (36) (constraints (25)), we adopt a scenario based approach with sample average approximation. A full description of these methods is beyond the scope of this paper; for theoretical details refer to [37] and to [38,39] for applications in energy management problems.

In short, the scenario based approach consists of generating a sufficiently high number $K \in \mathbb{N}$ of variable realizations based on data collection statistics. Each variable realization is called *scenario* and gives rise to a deterministic set of constraints. The optimization is conducted by considering all the scenarios simultaneously and minimizing the overall constraints violation.

In view of this, modifications to some of the constraints and costs derived in Section 3 are needed.

The latter are provided in the following subsections.

Modified constraints for the scenario setting

The modifications to constraints (1)–(9) and (15)–(17) account for the inclusion of both deterministic requests and the additional stochastic set, i.e., $Q(S, F) \cup Q'(S, F)$. The resulting constraints are indicated in the following as (1')–(9') and (15')–(17'). Other constraints need to be properly designed to account for the stochastic setting. They are designed as follows.

For the EVs in $Q'(S, F)$, constraints (10) are removed. Given the realization i^k , we define an additional cost function to minimize allocation deviations

$$Q_{i^k}^a = \sum_{j \in R} \sum_{t=S}^F q_{i^k}^a(t) a_{ij}(t),$$

where the non negative weights $q_{i^k}^a(t)$ are defined such that $q_{i^k}^a(t) = 0$, $t \in [S, \alpha_{i^k} - 1] \cup [\delta_{i^k} + 1, F]$. A suitable choice for $q_{i^k}^a(t)$ is to assign increasing values as t is far from the interval $[\alpha_{i^k}, \delta_{i^k}]$, e.g., $q_{i^k}^a(t) = \alpha_{i^k} - t$, for $t \leq \alpha_{i^k}$ and $q_{i^k}^a(t) = t - \delta_{i^k}$, for $t \geq \delta_{i^k}$. From this, we define the average allocation cost across scenarios as

$$Q_i^a = \frac{1}{K} \sum_{k=1}^K Q_{i^k}^a,$$

and the total allocation cost over all stochastic vehicles

$$Q^a = \sum_{i \in Q'(S, F)} Q_i^a. \quad (37)$$

Constraint (11) result in

$$a_{ij}(t) - a_{ij}(t-1) \leq \zeta_{i^k, j}^{D, +}(t), \quad (38a)$$

$$a_{ij}(t-1) - a_{ij}(t) \leq \zeta_{i^k, j}^{D, -}(t), \quad (38b)$$

$$\zeta_{i^k, j}^{D, +}(t) \geq 0, \quad (38c)$$

$$\zeta_{i^k, j}^{D, -}(t) \geq 0, \quad (38d)$$

$\forall j \in R$ and $t \in [S+1, F]$ with $\zeta_{i^k, j}^{D, -}(t)$ and $\zeta_{i^k, j}^{D, +}(t)$ additional decision variables. This results in the deviation penalty cost

$$Q_{i^k}^D = \sum_{j \in R} \sum_{t=S+1}^F q_{i^k}^D(t) (\zeta_{i^k, j}^{D, -}(t) + \zeta_{i^k, j}^{D, +}(t)),$$

where $q_{i^k}^D(t) = 0$ for $t \notin D_{i^k}$ and strictly positive otherwise. The subsequent scenario-based cost function is then defined as

$$Q^D = \sum_{i \in Q'(S, F)} Q_i^D, \quad Q_i^D = \frac{1}{K} \sum_{k=1}^K Q_{i^k}^D. \quad (39)$$

Battery dynamics are modified, for those $i \in Q'(S, F)$, as

$$e_{ik}(t+1) = \lambda_i e_{ik}(t) + \tau \left[\eta_i^+ z_{i, \bullet}^+(t) - \frac{1}{\eta_i^-} z_{i, \bullet}^-(t) \right], \quad (40)$$

for $t \in [S, F-1]$ with $e_{ik}(\alpha_{i^k}) = e_{i^k, \alpha_{i^k}}$ as the initial condition.

The related constraints assume the form

$$e_{ik}(t) \geq e_i^{\min} - \zeta_{i^k}^{e, -}(t), \quad (41a)$$

$$e_{ik}(t) \leq e_i^{\max} + \zeta_{i^k}^{e, +}(t), \quad (41b)$$

$$\zeta_{i^k}^{e, +}(t) \geq 0, \quad (41c)$$

$$\zeta_{i^k}^{e, -}(t) \geq 0, \quad (41d)$$

$\forall t \in [S, F]$ and

$$e_{ik}(\alpha_{i^k}) \leq e_{i^k, \alpha_{i^k}} + y_{\alpha_{i^k}}^+, \quad (42a)$$

$$e_{ik}(\alpha_{i^k}) \geq e_{i^k, \alpha_{i^k}} - y_{\alpha_{i^k}}^-, \quad (42b)$$

$$y_{\alpha_{i^k}}^+ \geq 0, \quad (42c)$$

$$y_{\alpha_{i^k}}^- \geq 0. \quad (42d)$$

$$e_{ik}(\delta_{i^k}) \leq e_{i^k, \delta_{i^k}}^+ + y_{\delta_{i^k}}^+, \quad (42e)$$

$$e_{ik}(\delta_{i^k}) \geq e_{i^k, \delta_{i^k}}^- - y_{\delta_{i^k}}^-, \quad (42f)$$

$$y_{\delta_{i^k}}^+ \geq 0, \quad (42g)$$

$$y_{\delta_{i^k}}^- \geq 0. \quad (42h)$$

Constraints in (41) are exploited to penalize the SOC deviations from the given limits. To do so, the following cost functions are introduced:

$$Q_{i^k}^e = \sum_{t=S}^F q_i^e(\zeta_{i^k}^{e, -}(t) + \zeta_{i^k}^{e, +}(t)),$$

with q_i^e positive weight that does not depend on the particular realization (conversely to (37) and (39)) and

$$Q_i^e = \frac{1}{K} \sum_{k=1}^K Q_{i^k}^e, \quad Q^e = \sum_{i \in Q'(S, F)} Q_i^e. \quad (43)$$

Constraints in (42) are instead used to penalize the violation on the initial and desired final SOC. To do so, the following cost are defined:

$$Q_i^a = \frac{1}{K} \sum_{k=1}^K q_i^a(y_{\alpha_{i^k}}^- + y_{\alpha_{i^k}}^+), \quad Q_i^d = \frac{1}{K} \sum_{k=1}^K q_i^d(y_{\delta_{i^k}}^- + y_{\delta_{i^k}}^+),$$

with q_i^a and q_i^d positive weights and

$$Q^a = \sum_{i \in Q'(S, F)} Q_i^a, \quad Q^d = \sum_{i \in Q'(S, F)} Q_i^d. \quad (44)$$

When considering the power exchanged with the grid, since the TSO service signals are not known and, therefore, modeled in a stochastic way, Eq. (22) are replaced by

$$P_{asm, k}^+(t) = z_{asm}^+(t) \omega_k^+(t), \quad (45a)$$

$$P_{asm, k}^-(t) = z_{asm}^-(t) \omega_k^-(t), \quad (45b)$$

where the subscript k denotes the k -th realization scenario. Consequently, constraint (24) is replaced by the following relaxed one:

$$\begin{aligned} & \sum_{i \in A(S, F) \cup Q(S, F) \cup Q'(S, F)} \left[z_{i, \bullet}^+(t) - z_{i, \bullet}^-(t) \right] - P_{idm}^+(t) - P_{idm}^-(t) + P_{dam}^-(t) \\ & + P_{dam}^-(t) - P_{asm, k}^+(t) + P_{asm, k}^-(t) \leq \zeta_k^{bal, +}(t), \end{aligned} \quad (46a)$$

$$\begin{aligned} & \sum_{i \in A(S, F) \cup Q(S, F) \cup Q'(S, F)} \left[z_{i, \bullet}^+(t) - z_{i, \bullet}^-(t) \right] - P_{idm}^+(t) - P_{idm}^-(t) + P_{dam}^-(t) \\ & + P_{dam}^-(t) - P_{asm, k}^+(t) + P_{asm, k}^-(t) \geq -\zeta_k^{bal, -}(t), \end{aligned} \quad (46b)$$

$$P_{\text{idm}}^+(t) + P_{\text{dam}}^+(t) + P_{\text{asm},k}^+(t) \leq P^{+, \max} + \zeta_k^{\text{buy},+}(t) \quad (46c)$$

$$P_{\text{idm}}^+(t) + P_{\text{dam}}^+(t) + P_{\text{asm},k}^+(t) \geq -\zeta_k^{\text{buy},-}(t) \quad (46d)$$

$$P_{\text{idm}}^-(t) + P_{\text{dam}}^-(t) + P_{\text{asm},k}^-(t) \leq P^{-, \max} + \zeta_k^{\text{sell},+}(t) \quad (46e)$$

$$P_{\text{idm}}^-(t) + P_{\text{dam}}^-(t) + P_{\text{asm},k}^-(t) \geq -\zeta_k^{\text{sell},-}(t) \quad (46f)$$

$$\zeta_k^{\text{bal},+}(t) \geq 0, \quad (46g)$$

$$\zeta_k^{\text{bal},-}(t) \geq 0, \quad (46h)$$

$$\zeta_k^{\text{buy},+}(t) \geq 0, \quad (46i)$$

$$\zeta_k^{\text{buy},-}(t) \geq 0, \quad (46j)$$

$$\zeta_k^{\text{sell},+}(t) \geq 0, \quad (46k)$$

$$\zeta_k^{\text{sell},-}(t) \geq 0, \quad (46l)$$

$\forall t \in [S, F]$. To the above constraints we associate the following cost function so as to minimize power imbalance on the scenarios:

$$Q_k^P = q^P \sum_{t=S}^F \left(\zeta_k^{\text{bal},-}(t) + \zeta_k^{\text{bal},+}(t) + \zeta_k^{\text{buy},-}(t) + \zeta_k^{\text{buy},+}(t) + \zeta_k^{\text{sell},-}(t) + \zeta_k^{\text{sell},+}(t) \right),$$

with q^P positive weight and

$$Q^P = \frac{1}{K} \sum_{k=1}^K Q_k^P. \quad (47)$$

Modified costs for the scenario setting

Similarly to the modifications made to the constraints, certain cost functions also need to be adapted for the scenario-based setting. In particular, in (28) we need to account for $\mathcal{A}(S, F) \cup \mathcal{Q}(S, F) \cup \mathcal{Q}'(S, F)$ instead of $\mathcal{A}(S, F) \cup \mathcal{Q}(S, F)$, thus resulting in (28)'. Coherently, the cost will be denoted as $C'_v(t)$. Furthermore the cost (31) is replaced with

$$C'_{\text{asm}}(t) = c_{\text{asm}}^+(t) \frac{\tau}{K} \sum_{k=1}^K P_{\text{asm},k}^+(t) - c_{\text{asm}}^-(t) \frac{\tau}{K} \sum_{k=1}^K P_{\text{asm},k}^-(t). \quad (48)$$

Finally the cost function (32) is modified to incorporate terms related to the scenario-based setting. The updated cost function, denoted as C'_y , is expressed as

$$C'_y = \sum_{i \in \mathcal{Q}(S, F)} w_{y,i}^+ y_{\delta,i}^+ + \sum_{i \in \mathcal{Q}(S, F)} w_{y,i}^- y_{\delta,i}^- + Q^\delta. \quad (49)$$

This adjustment accounts for additional stochastic considerations in the allocation process.

Optimal problem for the scenario setting

When expanding the optimization framework to include stochastic requests $\mathcal{Q}'(S, F)$ alongside the already allocated vehicles $\mathcal{A}(S, F)$ and the deterministic requests $\mathcal{Q}(S, F)$, the former overall cost J_c in (33) is substituted with

$$J'_c = \iota_C \left[\sum_{t=S}^F [C'_v(t) + C_{\text{idm}}(t) + C_{\text{dam}}(t) + C'_{\text{asm}}(t)] + C'_y \right] + \iota_Q [Q^a + Q^D + Q^e + Q^a + Q^P]. \quad (50)$$

The above cost takes into account both the modified expenses due to the scenario setting and the costs related to the constraint violations.

The scalars $\iota_C, \iota_Q \geq 0$ provide adjustable weighting factors, allowing the optimization to balance cost minimization and constraint adherence. The final optimization problem is formulated as follows:

$$\min J'_c$$

s.t.

Power variables, power rate, one station assignment (1)' – (9)',

Arrival and departure time for deterministic vehicles only (10),

Arrival and departure time for stochastic vehicles only (38),

Unavailability for deterministic vehicles only (11),

Unavailability for stochastic vehicles only (39),

Already assigned vehicles (8),

SOC for deterministic vehicles only (12) – (14),

SOC for stochastic vehicles only (40) – (42),

charging station constraints (15)' – (17)',

The idm and dam variables (18),

Ancillary service market variables (19) – (21), (23),

Ancillary service market power (45),

Power balance (46).

(51)

A notable exploitation of the scenario approach we just introduced is when only prediction data are available, e.g., the day (or any other time period) before. In such a case, the scenario approach can be used for an offline solution computation upon which the online algorithm will later work. A visual description of the runtime management with the scenario approach is reported in Fig. 2. Problem (51) can be conveniently exploited to compute the $\bar{P}_{\text{idm}}^+(t)$, $\bar{P}_{\text{idm}}^-(t)$, $\bar{s}^-(t)$, $\bar{s}^+(t)$ and $\bar{m}(t)$ to be used in the constraints (25) for running Algorithm 1.

These values are computed offline, the day before real-time scheduling begins, using a historical data-driven approach. Since no booking requests have been collected at that point, both $\mathcal{Q}(S, F)$ (incoming deterministic requests) and $\mathcal{A}(S, F)$ (already allocated vehicles) are empty, while all expected future EV arrivals are treated stochastically and included in $\mathcal{Q}'(S, F)$.

Such so defined offline algorithm runs (51) and not only provides $\bar{P}_{\text{idm}}^+(t)$, $\bar{P}_{\text{idm}}^-(t)$, $\bar{s}^-(t)$, $\bar{s}^+(t)$ and $\bar{m}(t)$, but it is also useful in deriving, given the $a_{ij}(t)$, the trajectory $w_{ij}(t)$ to be used in problem (35) of Algorithm 1. It is important to highlight that problem (35) serves as an approximation to account for scenario realizations when handling real-time EV allocations.

A more precise, but computationally more expensive, alternative would be to replace Algorithm 1 with Algorithm 2, which integrates the deterministic contributions from incoming vehicles in $\mathcal{Q}(S, F)$, the already allocated vehicles in $\mathcal{A}(S, F)$, and the stochastic impact of future incoming requests in $\mathcal{Q}'(S, F)$. Similar to Algorithm 1, Algorithm 2 is executed whenever the queue C is not empty, meaning at least one new allocation request is pending.

6. Numerical examples

In this section, the proposed online algorithm is evaluated through numerical simulations covering a range of test cases, from small-scale to large-scale scenarios. Additionally, a comparative analysis is conducted against a greedy allocation policy with optimized power profiles to benchmark its performance. The task of the online management algorithm (see Section 4) is to optimally handle deviations from this precomputed day-ahead plan as they occur in real-time and dynamically reschedule power profiles as needed. To test the algorithm we suppose

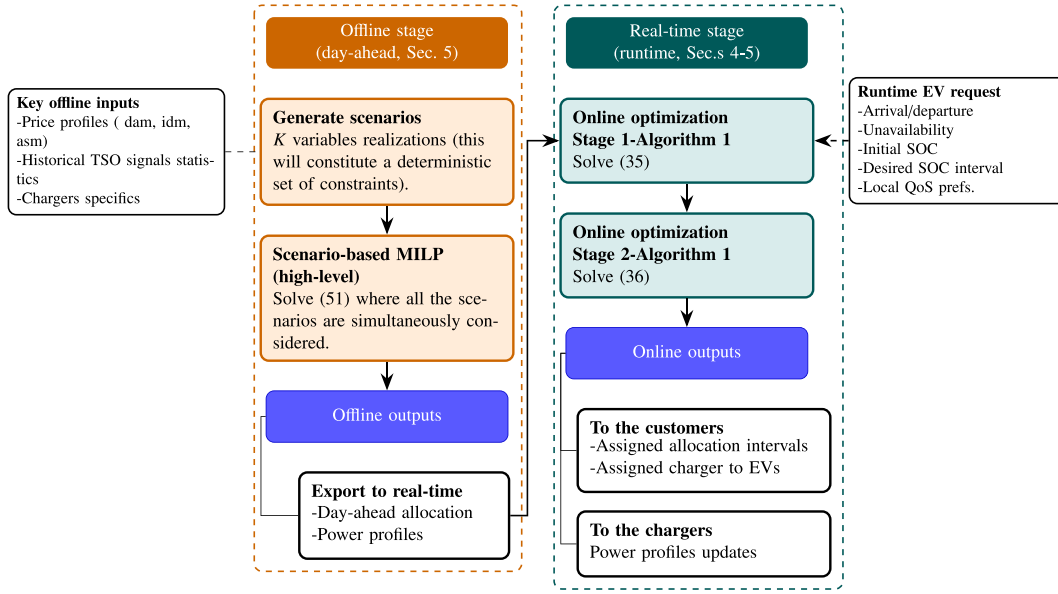


Fig. 2. Runtime management with scenario approach.

Algorithm 2: Parking manager with possible stochastic data.

Input: $C, S \in \mathbb{R}, F \in \mathbb{R}$
Output: $\mathcal{A}(S, F), \mathcal{Q}(S, F), \bar{\alpha}_i, \bar{\delta}_i$
 $\mathcal{Q}(S, F) \leftarrow \emptyset;$
for $i \in C$ **do**
 $\mathcal{Q}(S, F) \leftarrow \mathcal{Q}(S, F) \cup \{i\};$
 $C \leftarrow C \setminus \{i\};$
 //solve the optimization problem where
 $\bar{P}_{idm}^+(t), \bar{P}_{idm}^-(t), \bar{s}^-(t), \bar{s}^+(t)$ and $\bar{m}(t)$ are provided by the
 offline algorithm;
 Solve (51) with the additional constraint (25);
 if (51) with (25) is feasible **then**
 the optimization problem provides $[\bar{\alpha}_i, \bar{\delta}_i];$
 if EV i accepts $[\bar{\alpha}_i, \bar{\delta}_i]$ **then**
 $\mathcal{A}(S, F) \leftarrow \mathcal{A}(S, F) \cup \{i\};$
 end
 else
 Notify to the EV i the allocation fail;
 end
 $\mathcal{Q}(S, F) \leftarrow \mathcal{Q}(S, F) \setminus \{i\};$
end

that each EVs asking for allocation are allocated in case of solution (i.e., it does not refuse the proposed allocation).¹

Finally, a scenario-based framework is employed to generate offline day-ahead solutions that account for future incoming EVs.

6.1. Simulation setup

The cost profiles on the three energy markets are derived from the Italian market on a random day in 2024 and provided by the Italian government GME [36]. The charging station characteristics are chosen randomly according to the following criteria. The maximum power $z_{\bullet,j}^{+,max}$ each j -th station can deliver is uniformly randomly chosen in the

interval $[45, 75]$ kW, while the maximum power $z_{\bullet,j}^{-,max}$ each station can absorb is uniformly randomly chosen in the interval $[30, 60]$ kW. EVs are indexed in the order they submit their allocation requests, with EV 0 being the first. The EV characteristics are randomly generated, regardless of the parking lot size, as follows:

1. Maximum incoming and outgoing power ($z_{i,\bullet}^{+,max}, z_{i,\bullet}^{-,max}$) are uniformly randomly chosen in the intervals $[20, 22]$ kW and $[10, 20]$ kW, respectively;
2. Ramp rates (G_i^+, G_i^-) are uniformly randomly chosen in the interval $[0.5, 1.5]$ kW/min;
3. Arrival and departure times α_i and δ_i are randomly selected according to a normal distribution centred, respectively at 9.00 and 17.00, both with a standard deviation of 1.5 hours;
4. The probability of an unavailability interval (we consider only at most one unavailability interval per vehicle) is chosen to be 0.2. If present, such an interval has a length chosen randomly with a maximum length of the 15% of the working day associated with each EV owner. The beginning of the unavailability interval is randomly chosen within the working day of each EV owner;
5. Charging and discharging efficiency η_i^+, η_i^- are uniformly randomly chosen in the interval $[0.94, 0.99]$;
6. Maximum and minimum energy (e_i^{max}, e_i^{min}) are uniformly randomly chosen in the intervals $[30, 60]$ kWh and $[5, 10]$ kWh, respectively;
7. Initial charge e_{i,α_i} is uniformly randomly chosen in the interval $[10, 50]$ % of the e_i^{max} , while the final charge range required by each EV $[e_{i,\delta_i}^-, e_{i,\delta_i}^+]$, is randomly selected within the interval $[10, 50]$ % of e_i^{max} .

6.2. Online parking management for a small-scale scenario

We consider a scenario involving 10 EVs and 3 different charging stations with randomly generated characteristics. Station 0 can serve up to 2 EVs simultaneously, i.e., $N_0 = 2$, while stations 1 and 2 are limited to one EV each, i.e., $N_1 = 1$ and $N_2 = 1$. Once the 10 EVs are randomly generated and their day-ahead allocation and power profiles are computed, they are fed into the online management Algorithm 1.

To simulate real-world uncertainties, we introduce Gaussian noise to emulate deviations between planned and real-time data. Specifically, initial and desired SOC values are modified using a Gaussian distribution

¹ Notice that the possibility of refusing an allocation is related to the “industrialization” of the algorithm in a commercial app and does not affect the optimization problem.

with zero mean and 10% standard deviation; arrival and departure times are adjusted using a Gaussian distribution with zero mean and a standard deviation of 45 minutes; unavailability intervals are also shifted using a Gaussian distribution.

Figs. 3 and 4 depict the SOC and the allocations of all the EVs for the online case, respectively, while the power bought and sold on the three markets is reported in Fig. 5. As can be seen, the proposed management algorithm is able to allocate all EVs.

Figs. 6 and 7 report the comparison of the total allocation (that is, how many vehicles are allocated regardless of which charging station) and the total incoming and outgoing power. Despite significant perturbations between online inputs and the forecasts used to compute the day-ahead plan, the management algorithm closely adheres to the offline allocation time slots while adapting power profiles as necessary. A more detailed analysis with additional plots is provided in Appendix B.1.

6.3. Online parking management for large-scale scenarios

We evaluate the proposed strategy on a realistic large-scale case with 90 EVs requesting allocation and a corporate parking facility with 60 identical charging stations. Each station can serve a single EV at a time, which from a modeling perspective is equivalent to a unique station with up to 60 contemporary recharges. Fig. 8 reports the scheduled input and output power on the three markets; the proposed management reliably serves all online EV requests.

To demonstrate scalability, we also consider a larger test with 150 EVs and 70 charging stations. Fig. 9 shows the power purchased from and returned to the grid across the three markets for this scenario.

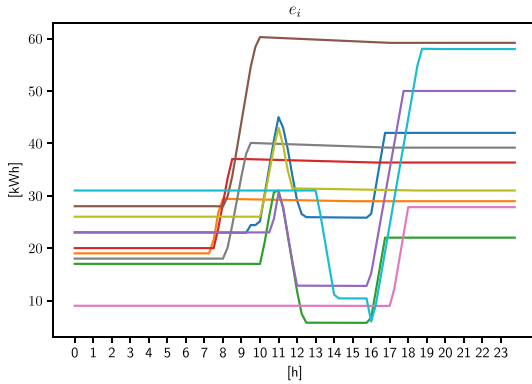


Fig. 3. SOC evolution of the 10 EVs.

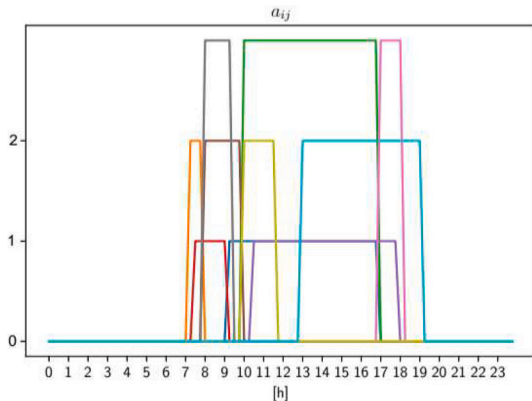
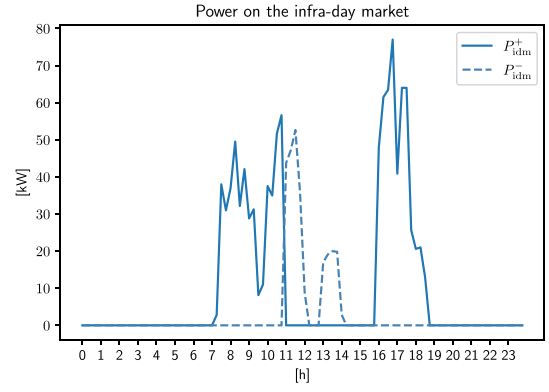
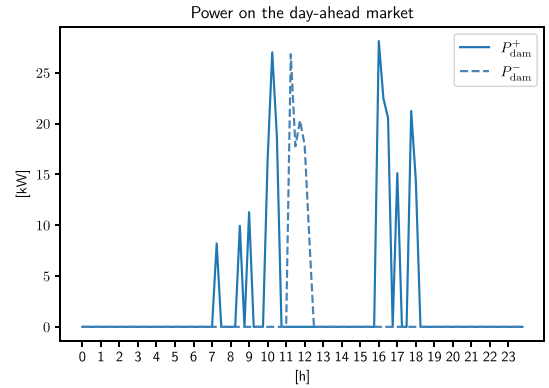


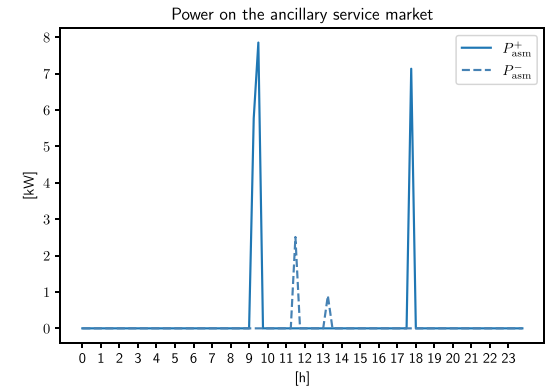
Fig. 4. Allocations of the 10 EVs to the charging stations (−1 denotes no allocation).



(a)



(b)



(c)

Fig. 5. Bought (+) and sold (−) powers on the three markets: (a) (idm); (b) (dam); (c) (asm) for the 10 EVs case.

We defer detailed comparisons with the day-ahead solution to Appendix B.2; despite online data perturbations, the algorithm preserves allocations and power profiles broadly consistent with the offline plan.

6.4. Scenario-based offline solution generation

We report a simulation for the scenario based approach developed in Section 5, for the large-scale scenario, i.e., 90 EVs and 60 equal charging stations. We consider the optimization problem (51) where all the

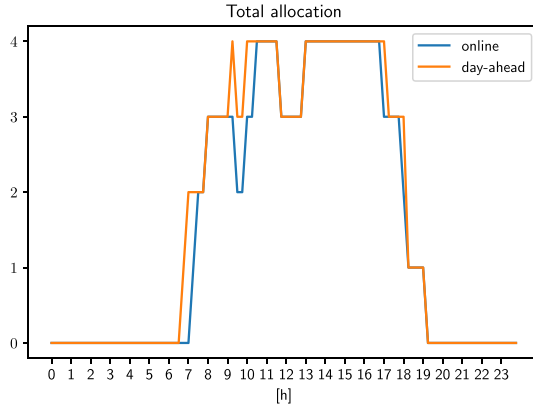


Fig. 6. Comparison between online (blue) and day-ahead computed (orange) total parking allocation for the 10 EVs case.

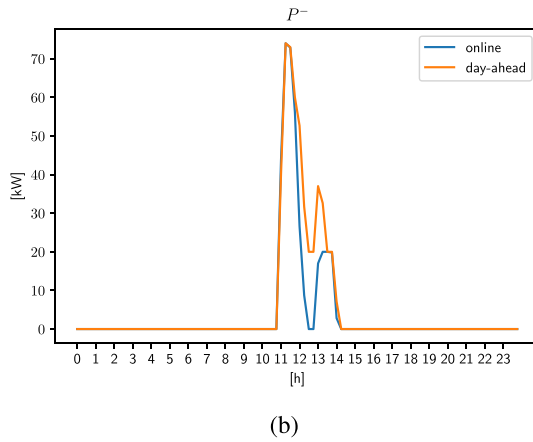
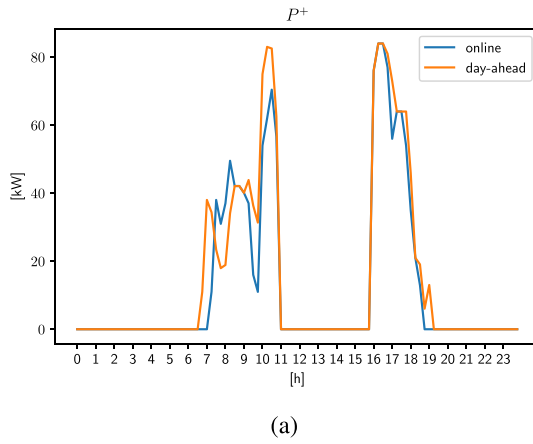


Fig. 7. Bought (a) and sold (b) total power comparison between online (blue) and day-ahead computed (orange) solutions for the 10 EVs case.

90 vehicles are involved, thus generating a day-ahead offline solution based on forecast data. The offline solution is computed prior to real-time operation and provides the reference trajectories that guide the online Algorithm 1.

The data of the EVs are generated randomly as already described at the beginning of this section. On such data, we construct ten scenarios and, as previously described, we optimize over all of them. More in detail, for each vehicle we randomly change the arrival and departure

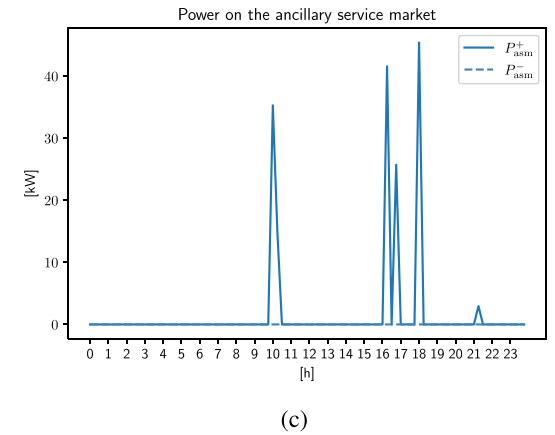
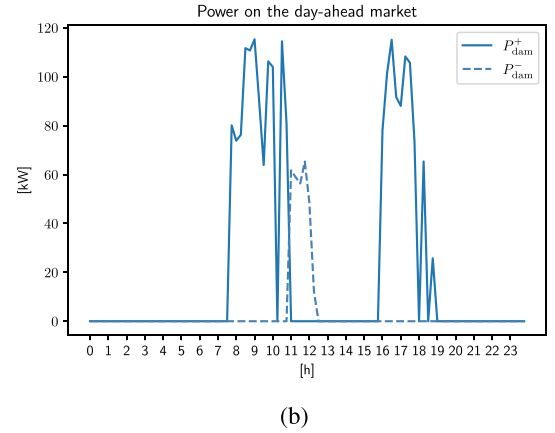
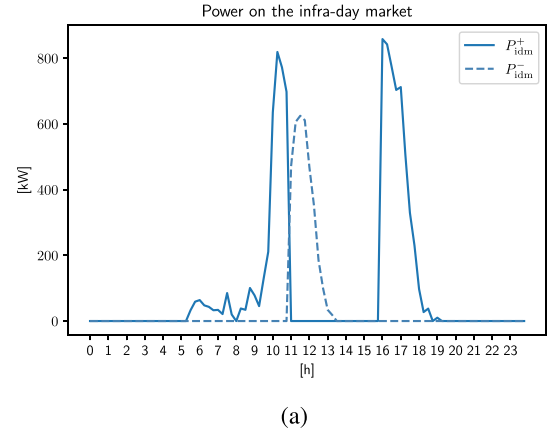
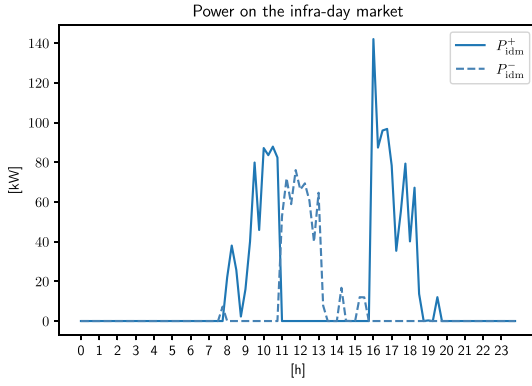
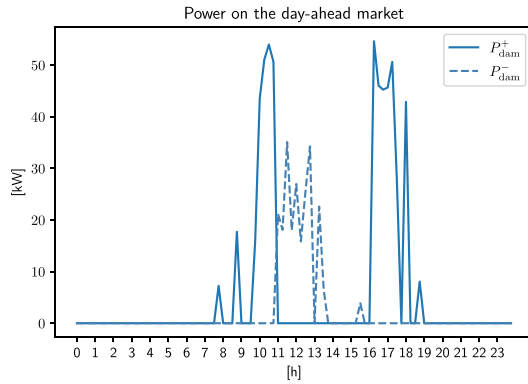


Fig. 8. Bought (+) and sold (−) powers on the three markets: ((a) idm); (b) (dam); (c) (asm) for the 90 EVs case.

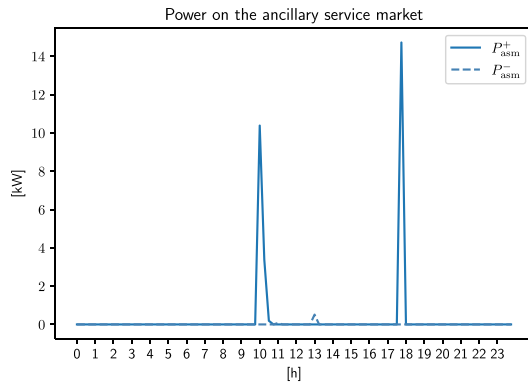
times with a zero mean Gaussian distribution of standard deviation of 45 minutes. The same standard deviation is used to randomly shift the unavailability interval. In addition to this, for each vehicle the initial SOC and final SOC interval boundaries are modified in the scenario generation through a zero mean Gaussian distribution with a 10% standard deviation. Finally, the downward and upward signals $\omega^-(t)$, $\omega^+(t)$ are generated randomly for the ten scenarios by modifying a given signal (we consider a historical trajectory as basis) with a zero mean and 10% standard deviation Gaussian distribution.



(a)



(b)



(c)

Fig. 9. Bought (+) and sold (−) powers on the three markets: (a) (idm); (b) (dam); (c) (asm) for the large 150 EVs case.

We deemed the ten scenarios generated in this way sufficiently rich and diverse to be adopted for optimization [51]. We selected ten as a pragmatic compromise between representativeness and offline tractability for our test-bed: the ensemble is sufficiently varied to produce day-ahead references averaged over historical statistics, while keeping the offline optimization computationally manageable. The solution of such problem provides an offline allocation of all the vehicles across all the scenarios.

In Fig. 10 we report the total allocation (in percentage) of the whole parking, while the incoming and outgoing power over the dam is reported in Fig. 11(a). Finally, Fig. 11(b) reports the average power

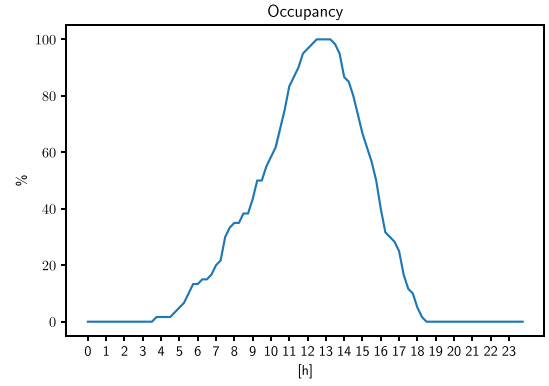
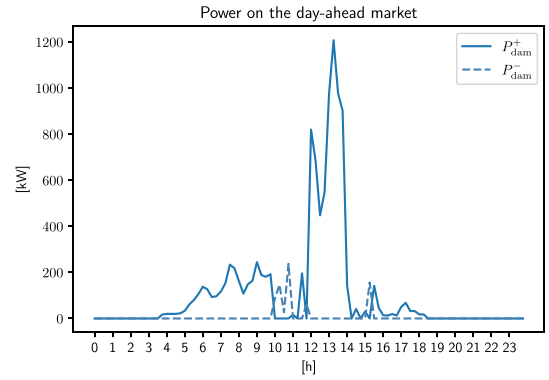
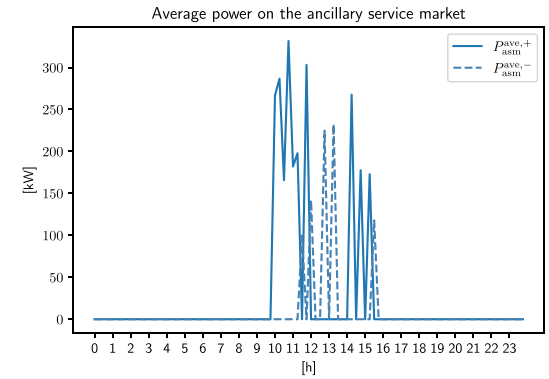


Fig. 10. Total parking percentage allocation for the 90 EVs scenario based case.



(a)



(b)

Fig. 11. Bought (+) and sold (−) powers for the 90 EVs scenario based case: (a) on the dam; (b) average power on the asm.

trajectory on the asm for the considered scenario, both incoming and outgoing.

To reduce the number of figures, we do not report the incoming and outgoing power profiles for the single vehicles. Nevertheless, we outline that, as explained in Section 5, the scenario solution can play the role of the day ahead solution that the management algorithm tries to adhere in real-time (that is a data for the online problem).

6.5. Simulation performance

The algorithms developed in this paper have been solved with Gurobi [40] and modelled with Pyomo [41], a Python based open source optimization modeling language. The optimization problems

were solved with Gurobi 11.0.0 via Pyomo. All the simulations are conducted on a 24 hours day with sampling time of 15 minutes, then the time instant $t \in \{0, 1, \dots, 95\}$. For readability, the results are reported in terms of corresponding hours.

The average runtime for online allocation is approximately 6 seconds per request on a Windows machine with an Intel i7 CPU and 16GB of RAM; allocation for the earliest requesting vehicle is typically fast (around 1s) and increases as additional vehicles request allocation.

Indeed, the online management algorithm computes an allocation whenever an EV requests one, while previously assigned allocations (for which owners have received parking/pickup times) remain unchanged. Each online instance introduces binary variables only for the requesting vehicle across the scheduling horizon, so the number of binary variables per instance is bounded by the time horizon and does not grow with fleet size. By contrast, the algorithm recomputes all power flows at each new request to optimize overall cost and satisfy new constraints, causing the number of continuous decision variables to linearly increase with the number of allocated EVs and charging stations. Consequently, computation time increases as the count of allocated vehicles grows although not dramatically since the number of binary variables remains stable. In particular, by referring to the last scheduling request, the small scenario with 10 EVs and 3 chargers, involves 123 continuous variables; the large-scale cases, require 21603 continuous variables for 90 EVs and 60 chargers, while 42003 continuous variables for 150 EVs and 70 chargers.

For the large parking facility cases we reported, the computation times increase from under one second to roughly 12s for the 90 EVs case, while they reach about 25s in the worst case (last allocations) for the 150 EVs case. These runtimes are compatible with online deployment even for large parking facilities. Moreover, the reported results were obtained on average performance hardware and with a code not optimized for a commercial deployment.

Additionally, an offline time period ahead strategy is computed by means of a scenario approach, as previously illustrated.

Specifically, 10 scenarios are generated offline based on empirically calibrated probability distributions. Differently from the online algorithm, where the number of binary variables is stable and refers, each time, to a single vehicle, the computation of each scenario involves all the EVs. It therefore requires longer computation times (up to roughly three hours for our cases), but remains an acceptable runtime considering that it generates an offline solution.

6.6. Comparison with the greedy policy

A comparative analysis is conducted here against the established greedy allocation strategy with optimized power flows to benchmark performance. More in detail, we compare the proposed allocation strategy for the large scale scenario (150 EVs and 70 charging stations) with a greedy baseline over a 15-day horizon (three working weeks, Monday-Friday). The greedy policy assigns an arriving EV to a free slot immediately or queues it until a slot becomes free; once assigned, the EV's charging power is optimized online as in the proposed strategy, i.e., power is continuously optimized according to new incoming vehicles to maximize the overall revenue and satisfy all vehicles constraints (battery dynamics, desired final SOC at the departure time, unavailability instants where the vehicle cannot be put in recharge/removed in/from a charging spot).

We evaluate both a scenario with final SOC flexibility and one without. Overall, across both scenarios, the proposed strategy is notably superior to the greedy policy. When no flexibility is allowed, the proposed optimal strategy yields higher revenues than the greedy policy and results in fewer vehicles left without an allocation, as shown in Table 2. For each vehicle, the customer dissatisfaction is defined, roughly speaking, as the percentage relative error between the final achieved SOC and the desired one (details and rigorous definitions are reported in Appendix B.3). Since no flexibility is considered, the customer dissatisfaction stems exclusively from unallocated vehicles.

Table 2

Statistical data over 15 days for the 150 EVs case with no flexibility.

	Optimal allocation		Greedy allocation	
	mean	std. deviation	mean	std. deviation
Tot. revenue	6396 €	513 €	5838 €	423 €
Tot. percentage of not allocation	1.33%	2.66%	24.89%	5.69%
Dissatisfaction	0.69%	5.95%	10.21%	18.76%

Table 3

Statistical data over 15 days for the 150 EVs case with flexibility.

	Optimal allocation		Greedy allocation	
	mean	std. deviation	mean	std. deviation
Tot. revenue	6481 €	447 €	6201 €	297 €
Tot. percentage of not allocation	0.00%	0.00%	23.55%	8.02%
Dissatisfaction	5.44%	5.99%	13.94%	17.12%

Allowing flexibility in the final SOC reduces the number of unallocated vehicles but increases aggregate dissatisfaction across the fleet, which in this case is not only related to the not allocated EVs. This effect is especially pronounced for the optimal strategy, while the greedy policy shows only modest change as shown in Table 3. As can be seen, the proposed optimal strategy “spreads” a small amount of dissatisfaction among many vehicles to increase the overall allocation (all of them are actually allocated as shown in Table 3).

We emphasize that allowing flexibility in the final SOC is an optional feature the parking manager may use to enable demand-response behaviour: customers could accept a slightly different final SOC in exchange for a discount on the final price. Several discount policies are in principle possible, but exploring them is beyond the scope of the paper. Here we adopt a simple symmetric discount mechanism and assume that EV owners always accept the proposed arrangement. Details on it, as well as additional details on the customer dissatisfaction metric, supplementary figures, and further technical specifications of the final SOC flexibility are provided in Appendix B.3.

7. Conclusions, limitations and future works

This paper proposes a novel parking management system for a company which offers employees the possibility of recharging their EVs. The designed management algorithm is specifically suited for runtime settings where charging occurs during work hours. Employees can book charging slots, specifying arrival and departure times, desired final charge levels, and unavailable time slots (e.g., due to meetings or other uninterruptible working activities).

The core of the system is a mixed integer linear program approach that optimizes charging schedules while accounting for various uncertainties, including arrival and departure times and initial SOC. The system supports the participation in energy markets and also accommodates a reduced QoS proposal when necessary. Furthermore it addresses real-world challenges, such as asynchronous booking requests and stochastic EVs behaviour. By optimizing resource allocation and energy management, the system ensures that each EV reaches its desired SOC by the end of the parking session while maximizing an objective function related to parking operations and energy market participation.

The proposed solution provides a comprehensive, adaptable, and practical framework for online EV charging scheduling in corporate parking lots. Also, it is amenable to deployment as a cloud service. Our approach handles both booking requests and forecasted data, enforces realistic human-centric constraints (desired SOC, time slot availability, immutability of assigned allocations while permitting power rescheduling), and supports market participation across dam, idm and asm horizons. It also accounts for infrastructure constraints, such as heterogeneous charging station capabilities and a limited number of charging

stations compared to the number of EVs requiring charging. Moreover, the proposed algorithm simultaneously solves a scheduling and power management problem, operating in a dynamic setting and being able to respond to new requests in real-time and replan accordingly.

Simulation results indicate higher request-satisfaction rates, higher operational profits, and bounded SOC deviations even under realistic disturbances of the offline plan. The proposed online algorithm is clearly superior to a baseline greedy allocation algorithm that employs optimized power profiles. Moreover, even when the input data are randomly generated with a significant standard deviation, the proposed algorithm outperforms the greedy allocation one by successfully allocating charging resources to nearly all the EVs asking for allocation and producing greater revenues together with considerably lower user dissatisfaction.

While the proposed algorithm is able to cope with significant deviations of online data with respect to the previsual offline one (thus demonstrating its practical applicability at runtime), it is not designed to cope with anomalous conditions such as sudden charging-station failures, communication outages and extreme price spikes. Future study will be extended along such directions to increase robustness and practical relevance. Scenarios might be enriched with explicit contingency realizations and anomalous events. As further research direction, we also plan to integrate on-site renewable generation and stationary storage models to jointly optimize V2G and local resources. At the moment such features are missing. Other interesting future research directions are related to test several flexibility pricing mechanisms against related user acceptance models (derived from on field test data) to find the most appropriate discount strategy for reduced QoS exploitation.

CRedit authorship contribution statement

Davide Liuzza: Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Software, Data curation, Conceptualization. **Elisa Mostacciolo:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Validation. **Valerio Mariani:** Writing – review & editing, Software, Formal analysis. **Luigi Iannelli:** Writing – original draft, Methodology, Conceptualization. **Francesco Vasca:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Giuseppe Silano:** Writing – review & editing, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Naturally avoiding contemporary in and out power flow

The following appendix shows that the optimization problem (36) prevents simultaneous inbound and outbound power flows.

Let us suppose by contradiction an optimal solution such that $P_{\text{idm}}^{+,*}(t) > 0$ and $P_{\text{idm}}^{-,*}(t) > 0$ at a certain time t . Problem (36) can be modified in an equivalent way by introducing the decision variable $P_{\text{idm}}^{+,*}(t) = P_{\text{idm}}^{+,*}(t) - P_{\text{idm}}^{-,*}(t)$ and modifying (24a) as

$$\sum_{i \in \mathcal{A}(S,F) \cup \mathcal{Q}(S,F)} \left[z_{i,*}^{+,*}(t) - z_{i,*}^{-,*}(t) \right] + P_{\text{idm}}^{+,*}(t) - P_{\text{dam}}^{+,*}(t) + P_{\text{dam}}^{-,*}(t) - P_{\text{asm}}^{+,*}(t) + P_{\text{asm}}^{-,*}(t) = 0. \quad (\text{A.1})$$

$\forall t \in [S, F]$.

We can show that there exists a feasible solution, denoted with $(\hat{\cdot})$, that keeps null one of the two between $P_{\text{idm}}^{+,*}(t)$ and $P_{\text{idm}}^{-,*}(t)$ yielding a cost value lower than the one provided by considering $P_{\text{idm}}^{+,*}(t), P_{\text{idm}}^{-,*}(t)$ both strictly positive.

Let us consider, without loss of generality, $P_{\text{idm}}^{-,*}(t) \leq 0$, then a possible feasible solution can be $\hat{P}_{\text{idm}}^{+,*}(t) = -P_{\text{idm}}^{-,*}(t)$ and $\hat{P}_{\text{idm}}^{-,*}(t) = 0$; (analogue considerations can be made if $P_{\text{idm}}^{+,*}(t) > 0$). The cost term $c_{\text{idm}}^{+,*} \tau \hat{P}_{\text{idm}}^{+,*}(t)$ provided by this solution is lower than the one provided by the $(*)$ solution. Indeed we can write the cost related to the $(*)$ solution as follows:

$$c_{\text{idm}}^{+,*} \tau P_{\text{idm}}^{+,*}(t) - c_{\text{idm}}^{-,*} \tau P_{\text{idm}}^{-,*}(t) = c_{\text{idm}}^{+,*} \tau [\hat{P}_{\text{idm}}^{+,*}(t) + P_{\text{idm}}^{-,*}(t)] - c_{\text{idm}}^{-,*} \tau P_{\text{idm}}^{-,*}(t).$$

This results in an additional positive cost $(c_{\text{idm}}^{+,*} - c_{\text{idm}}^{-,*}) \tau P_{\text{idm}}^{-,*}(t)$. Then, being the solution that keeps at the same time one of the two powers null feasible and better than the original one, the latter is not optimal. This contradicts its optimality hypothesis.

The exact same reasoning can be applied for the powers on the dam and asm or combinations among them. Again, by contradiction, let us suppose there exists an optimal solution with $z_{i,*}^{+,*}(t) > 0$ and $z_{i,*}^{-,*}(t) > 0$ for some i . Let us also introduce the new decision variable $u_i(t)$ with the constraint

$$u_i(t) = \eta_i^{+,*} z_{i,*}^{+,*}(t) - \frac{1}{\eta_i^{-,*}} z_{i,*}^{-,*}(t). \quad (\text{A.2})$$

The optimization problem can be equivalently written considering this additional decision variable, constraint (A.2) and modifying (12) as

$$e_i(t+1) = \lambda_i e_i(t) + \tau u_i(t), \quad (\text{A.3})$$

$\forall i \in \mathcal{A}(S,F) \cup \mathcal{Q}(S,F)$. If $u_i^{*}(t) > 0$ we can consider the candidate solution of the optimization problem that differs from the original one in having the new powers entering and leaving the i -th vehicle respectively as $\hat{z}_{i,*}^{+,*}(t) > 0$ and $\hat{z}_{i,*}^{-,*}(t) = 0$ such that $u_i^{*}(t) = \eta_i^{+,*} \hat{z}_{i,*}^{+,*}(t)$. Let us call $\hat{z}_{i,*}^{+,*}(t) = z_{i,*}^{+,*}(t) - \hat{z}_{i,*}^{+,*}(t)$. We obviously have that $\eta_i^{+,*} \hat{z}_{i,*}^{+,*}(t) = \frac{1}{\eta_i^{-,*}} z_{i,*}^{-,*}(t)$.

From this, it is possible to rewrite the cost contribution of the original solution $c_v^{-,*} \tau z_{i,*}^{-,*}(t) - c_v^{+,*} \tau z_{i,*}^{+,*}(t)$ as $c_v^{-,*} \tau z_{i,*}^{-,*}(t) - c_v^{+,*} \tau [\hat{z}_{i,*}^{+,*}(t) + \bar{z}_{i,*}^{+,*}(t)]$ which in turn is equal to $(c_v^{-,*} \eta_i^{-,*} - c_v^{+,*}) \tau \hat{z}_{i,*}^{+,*}(t) - c_v^{+,*} \tau \bar{z}_{i,*}^{+,*}(t)$. By remembering (27), the term $c_v^{-,*} \eta_i^{-,*} - c_v^{+,*}$ is strictly positive. Therefore, the cost associated with the original solution is higher than the one of the proposed candidate solution.

However, it is worth to highlight that the candidate solution does not respect constraint (24a). Indeed, the term $z_{i,*}^{+,*} - z_{i,*}^{-,*} = \hat{z}_{i,*}^{+,*} + \bar{z}_{i,*}^{+,*} - z_{i,*}^{-,*}$ is replaced by the sole term $\hat{z}_{i,*}^{+,*}$. The missing term $\bar{z}_{i,*}^{+,*} - z_{i,*}^{-,*}$ is a positive power entering into the vehicle that needs to be balanced for the new candidate solution via reducing the imported power from the three markets. This further translates in an additional reduction of the overall cost.

A similar reasoning can be done in case $u_i^{*}(t) \leq 0$. Indeed, the candidate solution $\hat{z}_{i,*}^{+,*}(t) = 0$ and $\hat{z}_{i,*}^{-,*}(t) \geq 0$ such that $u_i^{*}(t) = -\eta_i^{-,*} \hat{z}_{i,*}^{-,*}(t)$ yields to a cost $c_v^{-,*} \tau \hat{z}_{i,*}^{-,*}(t)$ which is lower than $c_v^{-,*} \tau \hat{z}_{i,*}^{-,*}(t) + (c_v^{-,*} \eta_i^{-,*} - c_v^{+,*}) \tau z_{i,*}^{+,*}(t)$ provided by the original solution. Also, to satisfy constraint (24a) the additional incoming power term $z_{i,*}^{+,*}(t) - \hat{z}_{i,*}^{-,*}(t)$ provided by the original solution must be compensated by a reduction in imported power, further lowering the overall cost.

Therefore, irrespective of the sign of $u_i^{*}(t)$, there always exists a solution with one of the two terms among $z_{i,*}^{+,*}(t), z_{i,*}^{-,*}(t)$ null and a reduced imported power which is feasible and provides a lower cost with respect to a solution that keeps both the terms $z_{i,*}^{+,*}(t), z_{i,*}^{-,*}(t)$ strictly positive. In view of this, the hypothesis of the latter being an optimal solution is contradicted.

Appendix B. Notes on numerical simulation

In this appendix, we provide additional details about the simulations performed and presented in Section 6.

B.1. Small-scale scenario

We highlight the ability of the proposed management algorithm for finding online solutions, by showing the comparison between online and

offline cases of two representative EVs, namely EV 0 and EV 4 (the first and the middle one requesting for allocation).

They are represented in Figs. B.12 and B.13, respectively, where the SOC (a), incoming $z_{i,*}^+$ (b) and outgoing $z_{i,*}^-$ (c) power profiles are shown.

By further analysing the behaviour of this single EV, we can see that our algorithm recomputes at any new allocation request issued by later EVs the power profiles for the EVs already allocated (indeed, the scheduled allocation does not change).

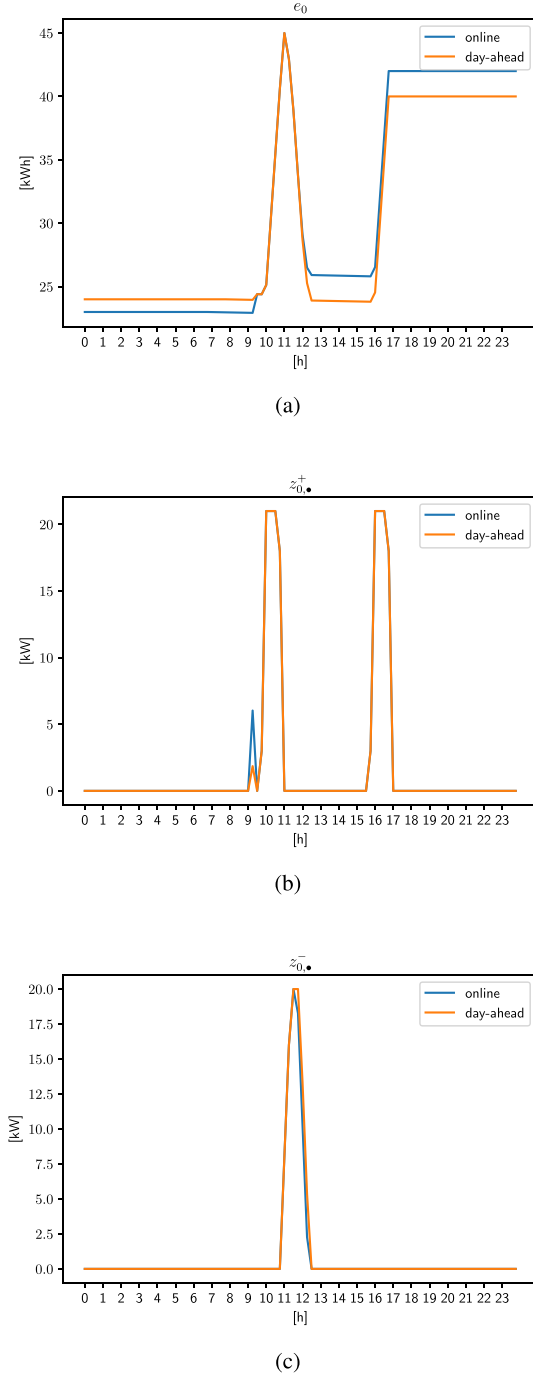


Fig. B.12. Comparison between the online (blue) and day-ahead (orange) solutions for EV 0 for the 10 EVs case: (a) SOC; (b) incoming power $z_{0,*}^+$; outgoing power $z_{0,*}^-$.

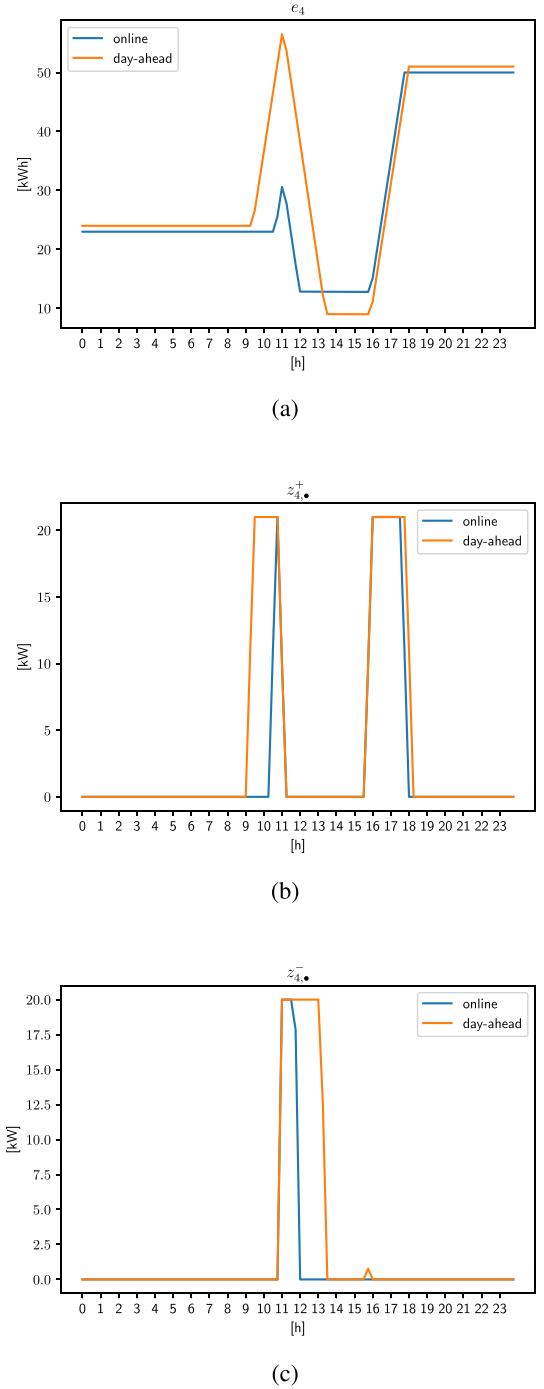


Fig. B.13. Comparison between the online (blue) and day-ahead (orange) solutions for EV 4 for the 10 EVs case: (a) SOC; (b) incoming power $z_{4,*}^+$; outgoing power $z_{4,*}^-$.

For the representative EV 0 and EV 4 this is shown in Figs. B.14 and B.15, respectively, where the subsequent SOC is shown. In particular, since EV 0 is the first one asking for allocation, it is subjected to nine other reschedules (each one triggered by a new incoming EV). The last reschedule (shot 9) is the one corresponding to the actual SOC profile (also reported in Fig. B.12(a) in blue) since no more re-computations will be performed.

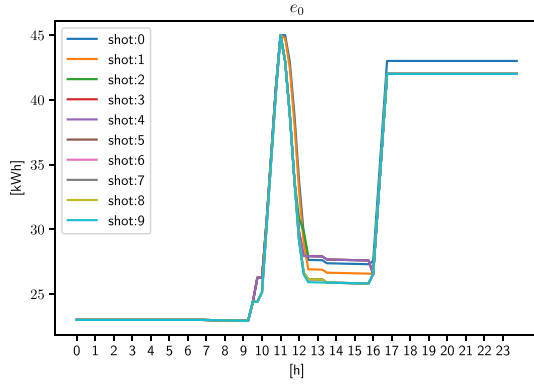
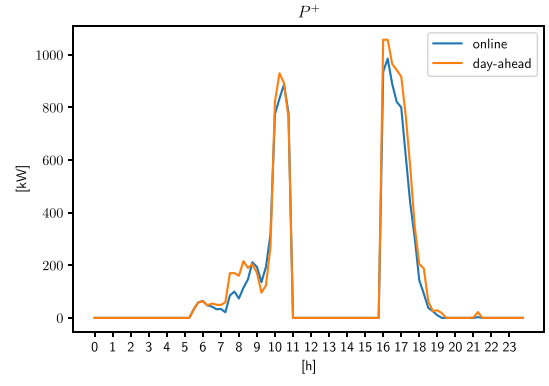


Fig. B.14. Online computations of the SOC solutions for EV 0 at any new allocation request for the 10 EVs case.



(a)

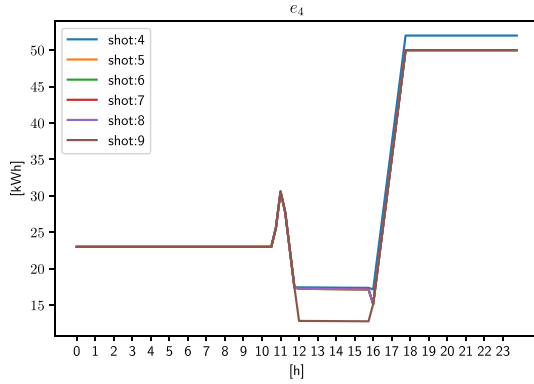
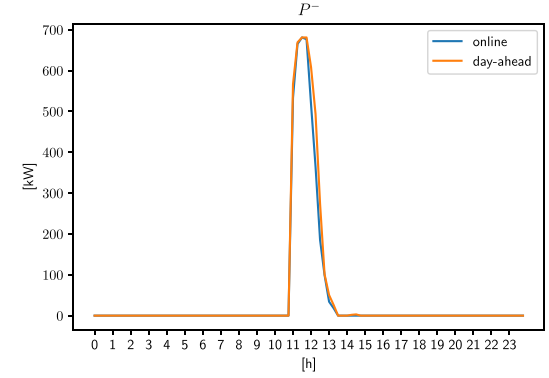


Fig. B.15. Online computations of the SOC solutions for EV 4 at any new allocation request for the 10 EVs case.



(b)

Fig. B.17. Bought (a) and sold (b) total power comparison between online (blue) and day-ahead computed (orange) solutions for the 90 EVs case.

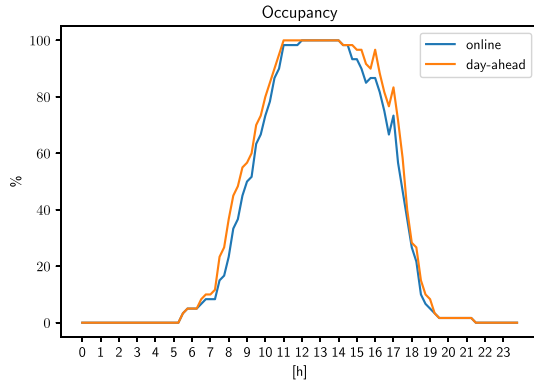


Fig. B.16. Comparison between online (blue) and day-ahead computed (orange) total parking percentage allocation for the 90 EVs case.

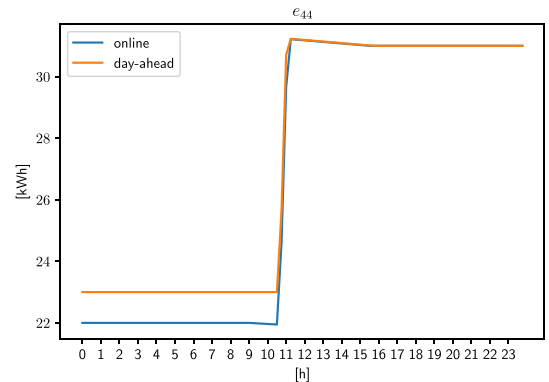


Fig. B.18. Comparison between the online (blue) and day-ahead computed (orange) SOC for EV 44 for the 90 EVs case.

With the same argument, being EV 4 the fifth vehicle asking for an allocation, it will be subjected to only five reschedules, as shown in Fig. B.15. Also in this case, the last reschedule (shot 9) corresponds to the actual SOC trajectory.

B.2. Large-scenarios

We compare the online solution with respect to a previously computed day-ahead solution on the nominal data. Despite the latter are subjected to significant perturbations on the online case, the management algorithm finds a solution whose allocations adhere reasonably to the day-ahead case. In Fig. B.16, the percentage of the total occupancy

for the large scale scenario with 90 EVs (100% corresponds to all the 60 lots simultaneously engaged) is compared, while in Fig. B.17 comparison of the total power is shown. We also detail the behaviour of one EV; see Fig. B.18 which represents the SOC behaviour. In particular the EV 44 has been taken as a representative example, being the one requiring allocation after half EVs.

Analogous results have been obtained also for the case of 150 EVs and 70 charging stations. Figs. B.19 and B.20 respectively report the parking allocation and the exchanged power comparison with respect to the offline solution.

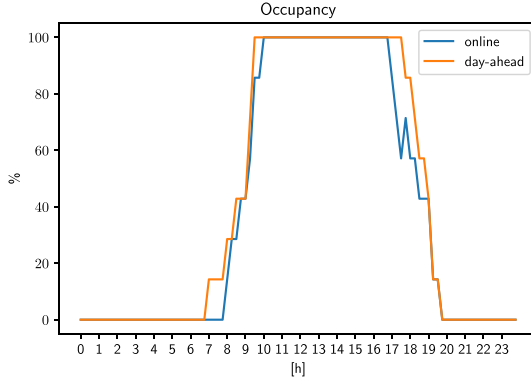
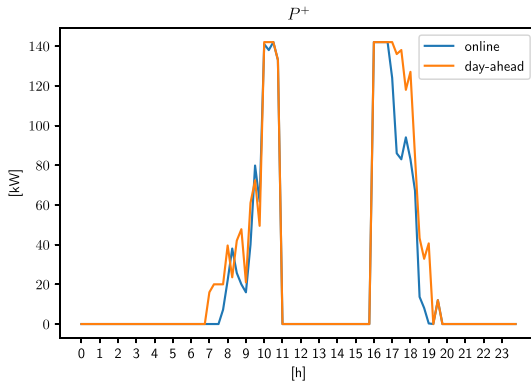
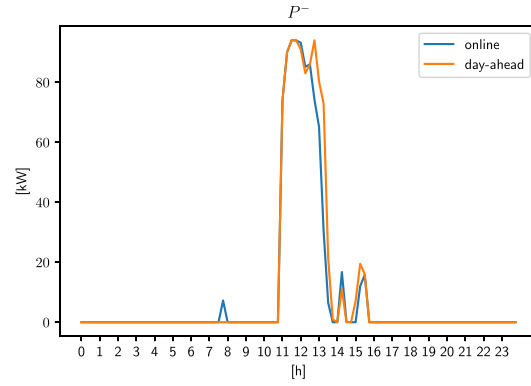


Fig. B.19. Comparison between online (blue) and day-ahead computed (orange) total parking percentage allocation for the large 150 EVs case.



(a)

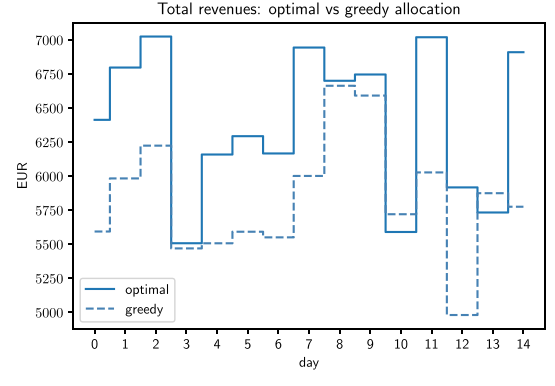


(b)

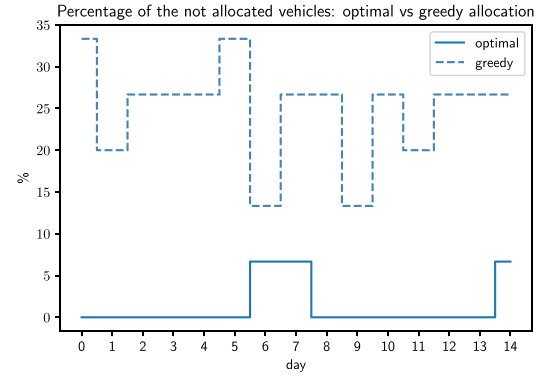
Fig. B.20. Bought (a) and sold (b) total power comparison between online (blue) and day-ahead computed (orange) solutions for the large 150 EVs case.

B.3. Comparison with greedy policy

The comparison examines two scenarios: one with no final SOC flexibility and one allowing limited SOC flexibility. When final SOC flexibility is disabled, the bounds values $y_{\delta,i}^{-,\max}$, $y_{\delta,i}^{+,\max}$ in constraints (14e), (14f) are set to zero, so every valid allocation must deliver the exact desired final SOC $[e_{i,\delta_i}^-, e_{i,\delta_i}^+]$. Fig. B.21 illustrates the comparison; Fig. B.22 shows the cumulative histogram of vehicles over the 15-days horizon, where the values on the vertical bars are the cumulative number of vehicles. The measure of the customer dissatisfaction



(a)



(b)

Fig. B.21. Comparison over 15 days for the 150 EVs case with not allowed flexibility in the final SOC: (a) total parking revenues; (b) percentage of not allocated vehicles.

is obtained according to the following percentage value

$$\text{diss}_i = \begin{cases} 100 \frac{e_{i,\delta_i}^- - e_i(\delta_i)}{e_{i,\delta_i}^-} & \text{if } e_i(\delta_i) < e_{i,\delta_i}^- \\ 100 \frac{e_i(\delta_i) - e_{i,\delta_i}^+}{e_{i,\delta_i}^+} & \text{if } e_i(\delta_i) > e_{i,\delta_i}^+ \\ 0 & \text{if } e_{i,\delta_i}^- \leq e_i(\delta_i) \leq e_{i,\delta_i}^+ \end{cases}$$

in case the i -th EV has been allocated, while

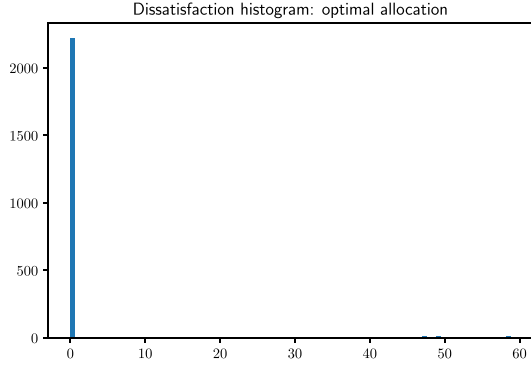
$$\text{diss}_i = 100 \frac{e_{i,\delta_i}^- - e_i(\alpha_i)}{e_{i,\delta_i}^-}$$

if no allocation is found for the i -th vehicle.

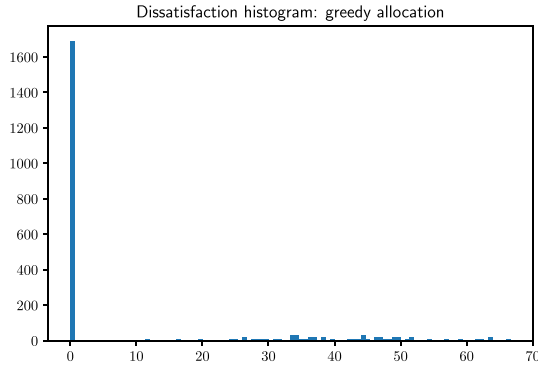
The proposed optimal strategy consistently outperforms the greedy policy by generating higher revenues and significantly reducing customer dissatisfaction. Although revenue variability is high, the optimal approach leverages the diversity in vehicle arrival and departure times to improve allocation efficiency. In contrast, the greedy strategy leads to a larger number of unallocated vehicles and, consequently, greater cumulative dissatisfaction, as shown in Fig. B.22.

In case with final SOC flexibility, the upward and downward flexibility bounds $y_{\delta,i}^{-,\max}$, $y_{\delta,i}^{+,\max}$ are randomly selected in the interval $[1, 5]$ kWh. We chose, therefore, to allow a modest amount of flexibility. The costs $w_{y,i}^+$, $w_{y,i}^-$ concurring in defining the penalty C_y provided in (32) have been set as

$$w_{y,i}^+ = w_{y,i}^- = \frac{1}{2} \left(\min_{t \in [S,F]} c_v^+(t) - \max_{t \in [S,F]} c_{\text{idm}}^+(t) \right)$$



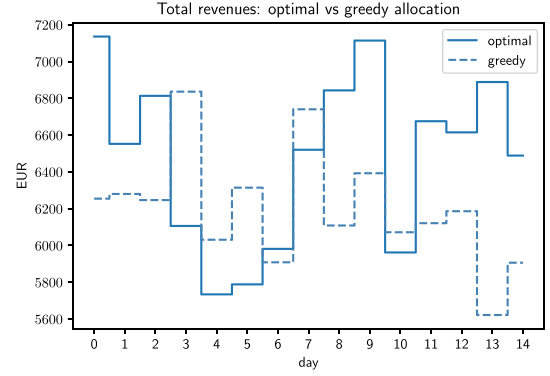
(a)



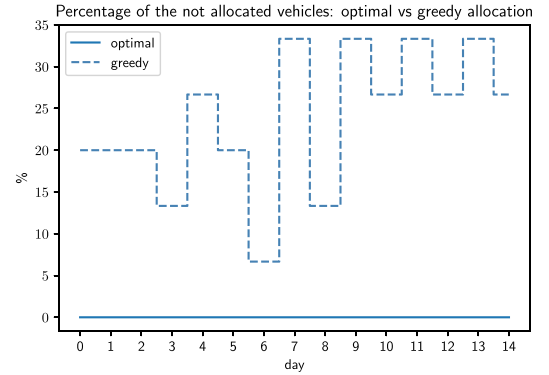
(b)

Fig. B.22. Cumulative dissatisfaction histogram over the 15 days horizon for the 150 EVs case with not allowed flexibility in the final SOC: (a) proposed optimal allocation strategy; (b) greedy allocation strategy.

for all vehicles $i \in \mathcal{V}$. This choice models a discount of a conservative lower bound on the manager's marginal profit. Indeed, suppose a vehicle is charged less than what is desired of the amount $y_{\delta,i}^-$ to allow the recharging of another vehicle. The manager will pay a cost $\omega_{y,i}^- y_{\delta,i}^-$ as a discount to the i -th vehicle for the reduced QoS. We chose such a discount the 50% of the lower bound $\min_{t \in [S,F]} c_v^+(t) - \max_{t \in [S,F]} c_{\text{idm}}^+(t)$ of the profit the parking manager can earn. The same amount is chosen, for keeping a simple symmetric structure, also for exceeding flexibility $y_{\delta,i}^+$.



(a)



(b)

Fig. B.23. Comparison over 15 days for the 150 EVs case with allowed flexibility in the final SOC: (a) total parking revenues; (b) percentage of not allocated vehicles.

Allowing modest SOC flexibility leads the proposed manager to exploit flexibility frequently, increasing total revenue and reducing the number of unallocated vehicles relative to the no-flexibility case. This can be interpreted with the fact that the management algorithm exploits the allowed flexibility “distributing” small amounts of dissatisfaction across the EVs to increase the number of allocated EVs and, consequently, its profit (despite the discount it has to pay). Fig. B.23 illustrates the comparison; Fig. B.24 shows the cumulative histogram.

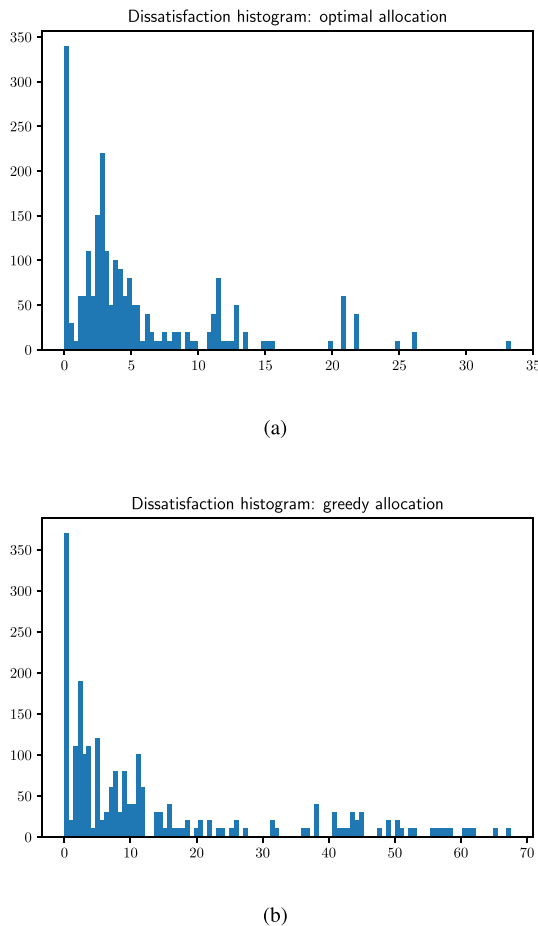


Fig. B.24. Cumulative dissatisfaction histogram over the 15 days horizon for the 150 EVs case with allowed flexibility in the final SOC: (a) proposed optimal allocation strategy; (b) greedy allocation strategy.

Data availability

Data will be made available on request.

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